

```

import numpy as np
import pandas as pd

import os
for dirname, _, filenames in os.walk('/kaggle/input'):
    for filename in filenames:
        print(os.path.join(dirname, filename))

import seaborn as sns
import matplotlib.pyplot as plt
import scipy.stats as st
%matplotlib inline

sns.set(style="whitegrid")

import warnings
warnings.filterwarnings('ignore')

df=pd.read_csv(r'C:\Users\HP\Downloads\6th- Advanced EDA project\6th-
Advanced EDA project\EDA- HEALTHCARE DOMAIN\heart.csv')

# print the shape
print('The shape of the dataset : ', df.shape)

```

The shape of the dataset : (303, 14)

```
df.head()
```

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	sl
0	63	1	3	145	233	1	0	150	0	2.3	0
1	37	1	2	130	250	0	1	187	0	3.5	0
2	41	0	1	130	204	0	0	172	0	1.4	2
3	56	1	1	120	236	0	1	178	0	0.8	2
4	57	0	0	120	354	0	1	163	1	0.6	2

```
df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 303 entries, 0 to 302
Data columns (total 14 columns):
#   Column      Non-Null Count  Dtype
---  -
0   age         303 non-null   int64
1   sex         303 non-null   int64
2   cp          303 non-null   int64
3   trestbps    303 non-null   int64

```

```

4   chol      303 non-null   int64
5   fbs       303 non-null   int64
6   restecg   303 non-null   int64
7   thalach   303 non-null   int64
8   exang     303 non-null   int64
9   oldpeak   303 non-null   float64
10  slope     303 non-null   int64
11  ca        303 non-null   int64
12  thal      303 non-null   int64
13  target    303 non-null   int64

```

```
dtypes: float64(1), int64(13)
```

```
memory usage: 33.3 KB
```

```
df.dtypes
```

```

age          int64
sex          int64
cp           int64
trestbps     int64
chol         int64
fbs          int64
restecg      int64
thalach      int64
exang        int64
oldpeak      float64
slope        int64
ca           int64
thal         int64
target       int64

```

```
dtype: object
```

```
df.describe()
```

	age	sex	cp	trestbps	chol	
count	303.000000	303.000000	303.000000	303.000000	303.000000	303.00
mean	54.366337	0.683168	0.966997	131.623762	246.264026	0.1485
std	9.082101	0.466011	1.032052	17.538143	51.830751	0.3561
min	29.000000	0.000000	0.000000	94.000000	126.000000	0.0000
25%	47.500000	0.000000	0.000000	120.000000	211.000000	0.0000
50%	55.000000	1.000000	1.000000	130.000000	240.000000	0.0000
75%	61.000000	1.000000	2.000000	140.000000	274.500000	0.0000
max	77.000000	1.000000	3.000000	200.000000	564.000000	1.0000

```
df.columns
```

```

Index(['age', 'sex', 'cp', 'trestbps', 'chol', 'fbs', 'restecg', 'thalach',
      'exang', 'oldpeak', 'slope', 'ca', 'thal', 'target'],
      dtype='object')

```

```
df['target'].unique()
```

```
array([1, 0])
```

```
df['target'].value_counts()
```

```
target
```

```
1    165
```

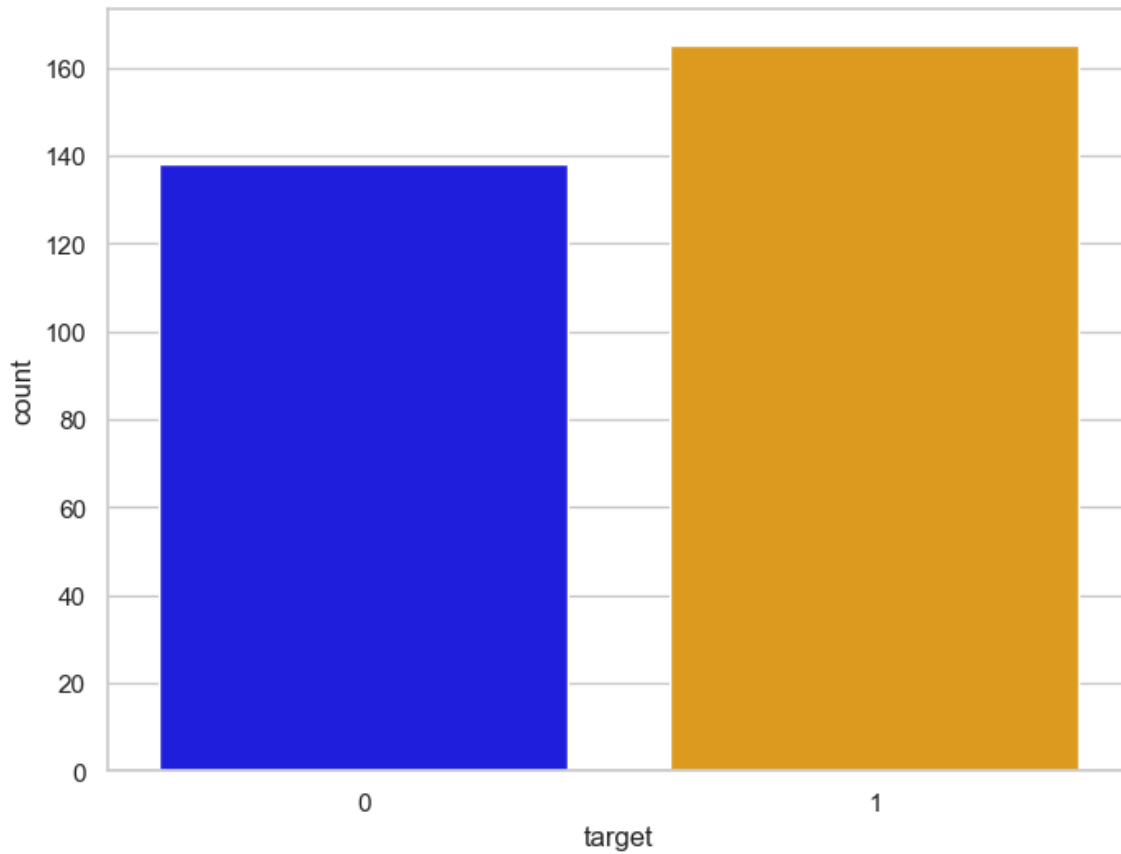
```
0    138
```

```
Name: count, dtype: int64
```

```
f,ax=plt.subplots(figsize=(8,6))
```

```
ax=sns.countplot(x='target',data=df,palette=['blue','orange'])
```

```
plt.show()
```



```
df.groupby('sex')['target'].value_counts()
```

```
sex  target
```

```
0    1         72
```

```
    0         24
```

```
1    0        114
```

```
    1         93
```

```
Name: count, dtype: int64
```

```
pd.crosstab(
```

```
df['sex'],
```

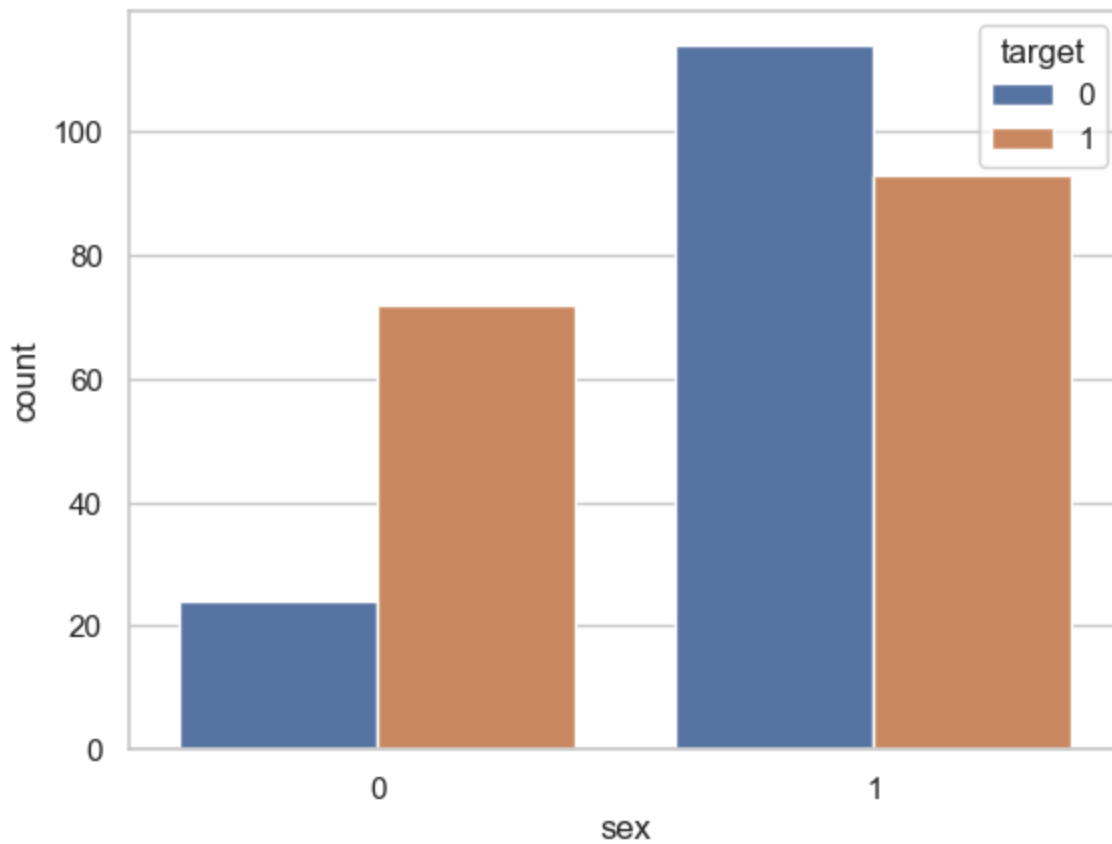
```
df['target'],
```

```
normalize='index'
```

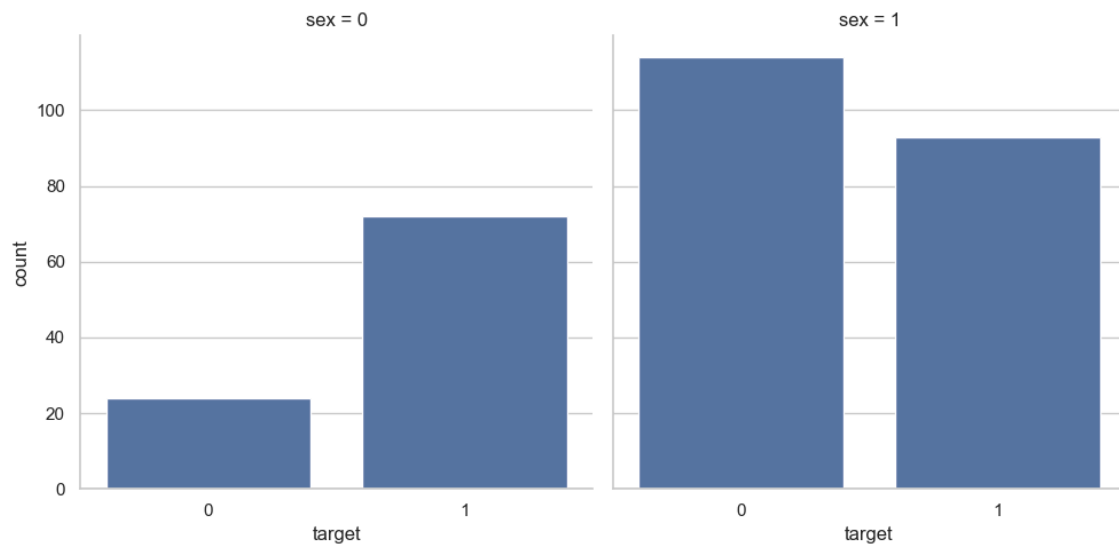
```
)*100
```

target	0	1	
sex			
0	25.000000	75.000000	
1	55.072464	44.927536	

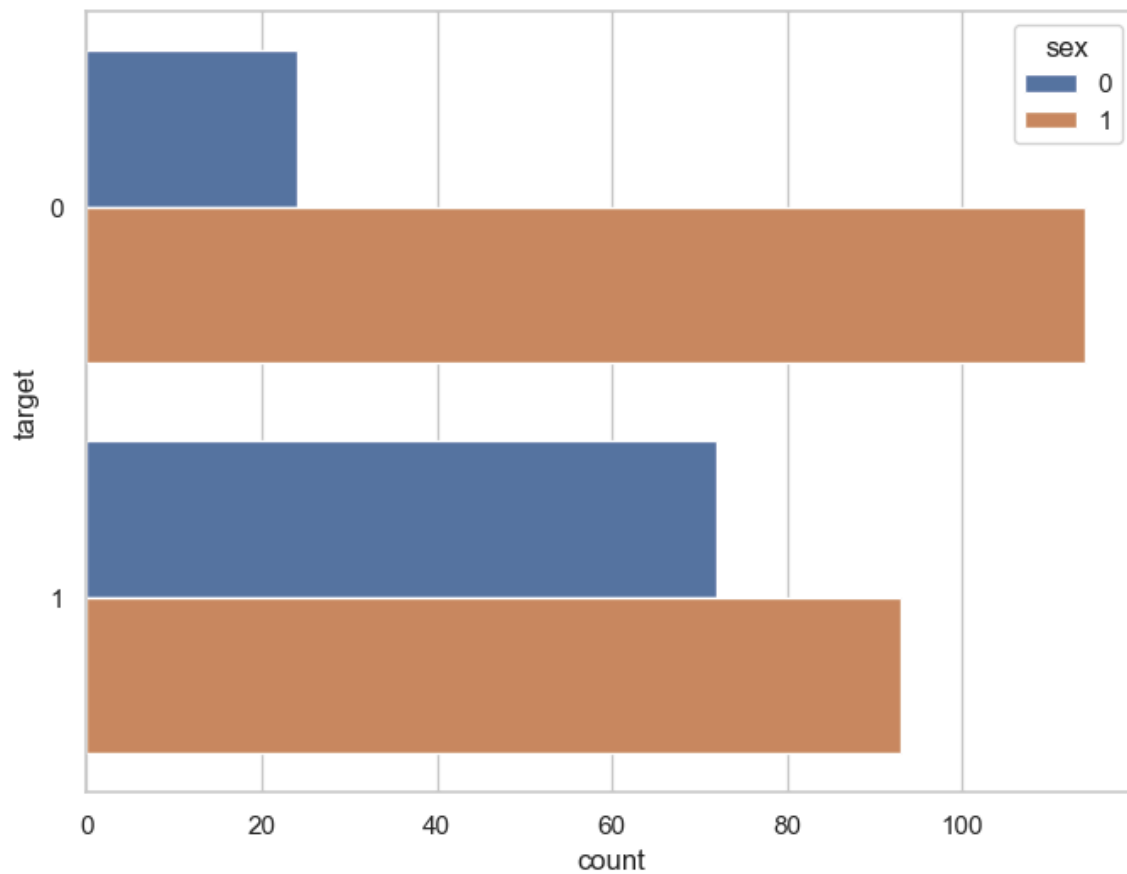
```
sns.countplot(
x='sex',
hue='target',
data=df
)
plt.show()
```



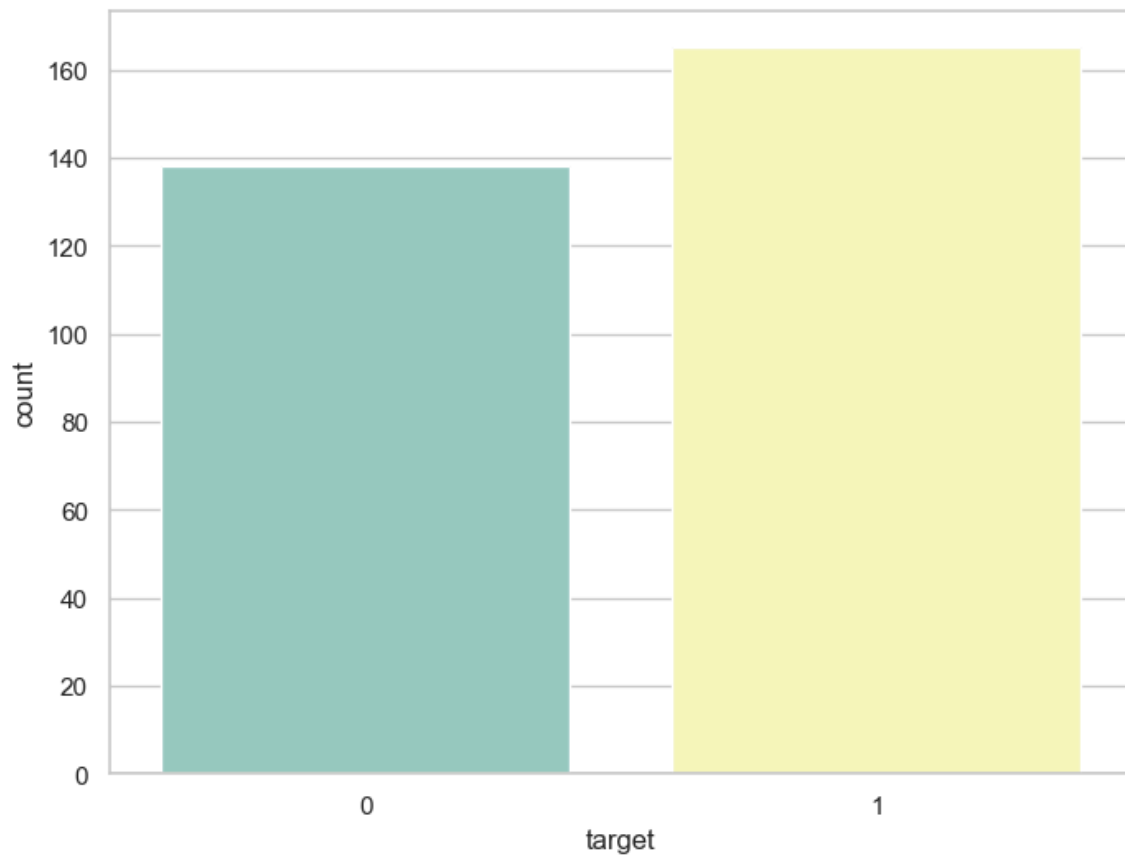
```
ax=sns.catplot(x='target',col='sex',data=df,kind='count',height=5,aspect=1)
```



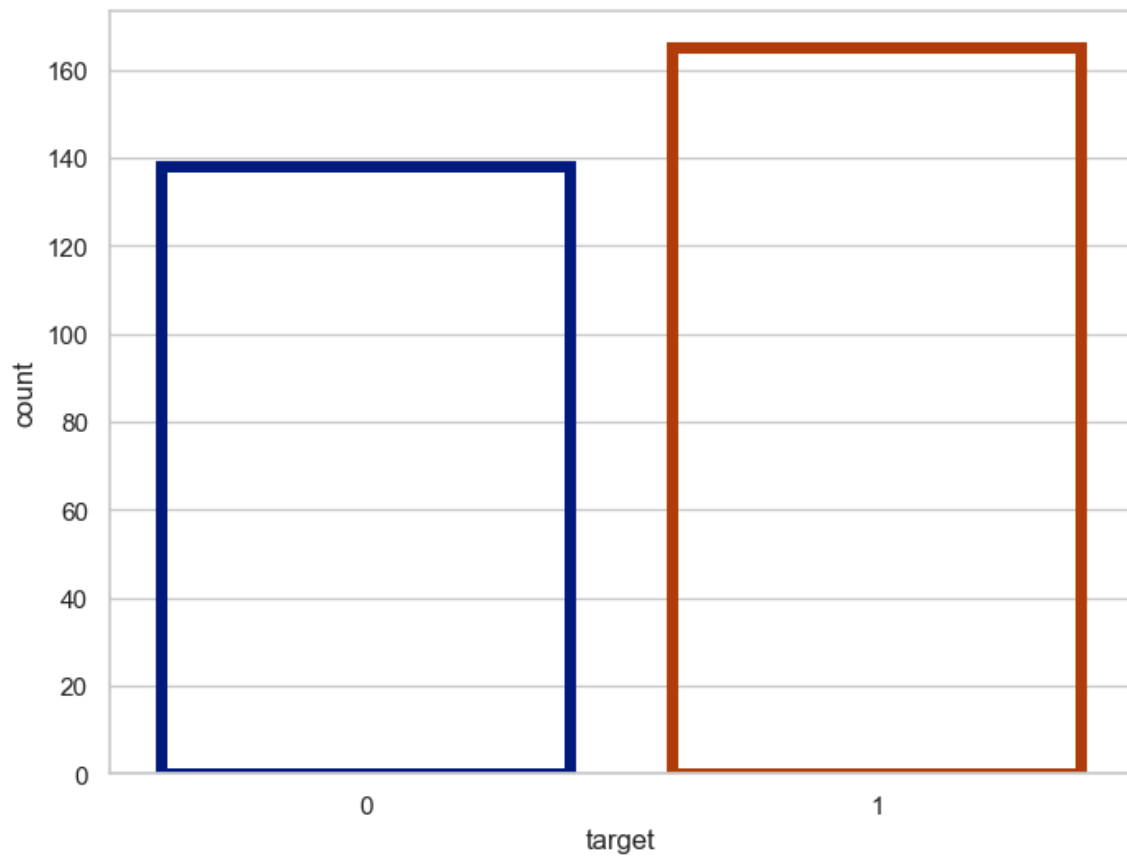
```
f,ax=plt.subplots(figsize=(8,6))
ax=sns.countplot(y='target',hue='sex',data=df)
plt.show()
```



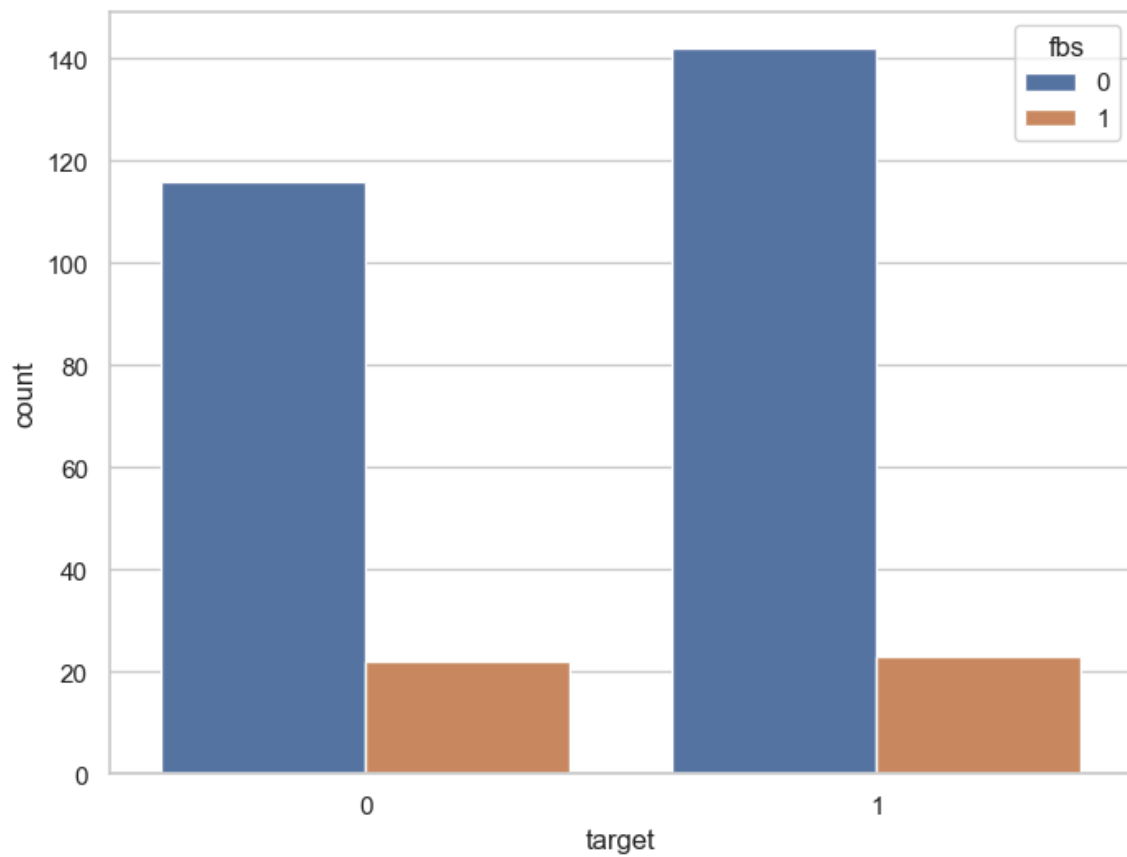
```
f, ax = plt.subplots(figsize=(8, 6))
ax = sns.countplot(x="target", data=df, palette="Set3")
plt.show()
```



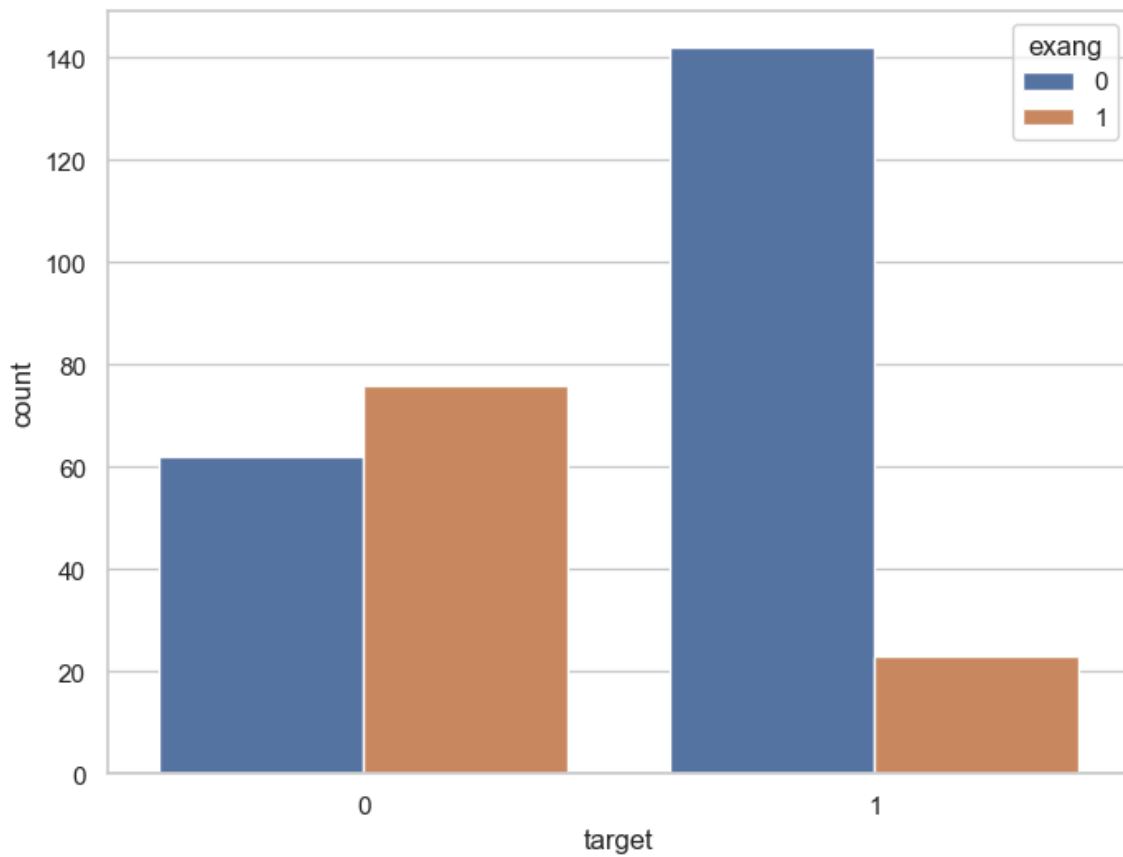
```
f, ax = plt.subplots(figsize=(8, 6))
ax = sns.countplot(x="target", data=df, facecolor=(0, 0, 0, 0), linewidth=5,
                  edgecolor=sns.color_palette("dark", 3))
plt.show()
```



```
f,ax=plt.subplots(figsize=(8,6))  
ax=sns.countplot(x='target',hue='fbs',data=df)  
plt.show()
```



```
f, ax = plt.subplots(figsize=(8, 6))
ax = sns.countplot(x="target", hue="exang", data=df)
plt.show()
```



```
correlation = df.corr()
```

```
correlation['target'].sort_values(ascending=False)
```

```
target      1.000000
cp          0.433798
thalach     0.421741
slope       0.345877
restecg     0.137230
fbs         -0.028046
chol        -0.085239
trestbps    -0.144931
age         -0.225439
sex         -0.280937
thal        -0.344029
ca          -0.391724
oldpeak     -0.430696
exang       -0.436757
Name: target, dtype: float64
```

```
df['cp'].nunique()
```

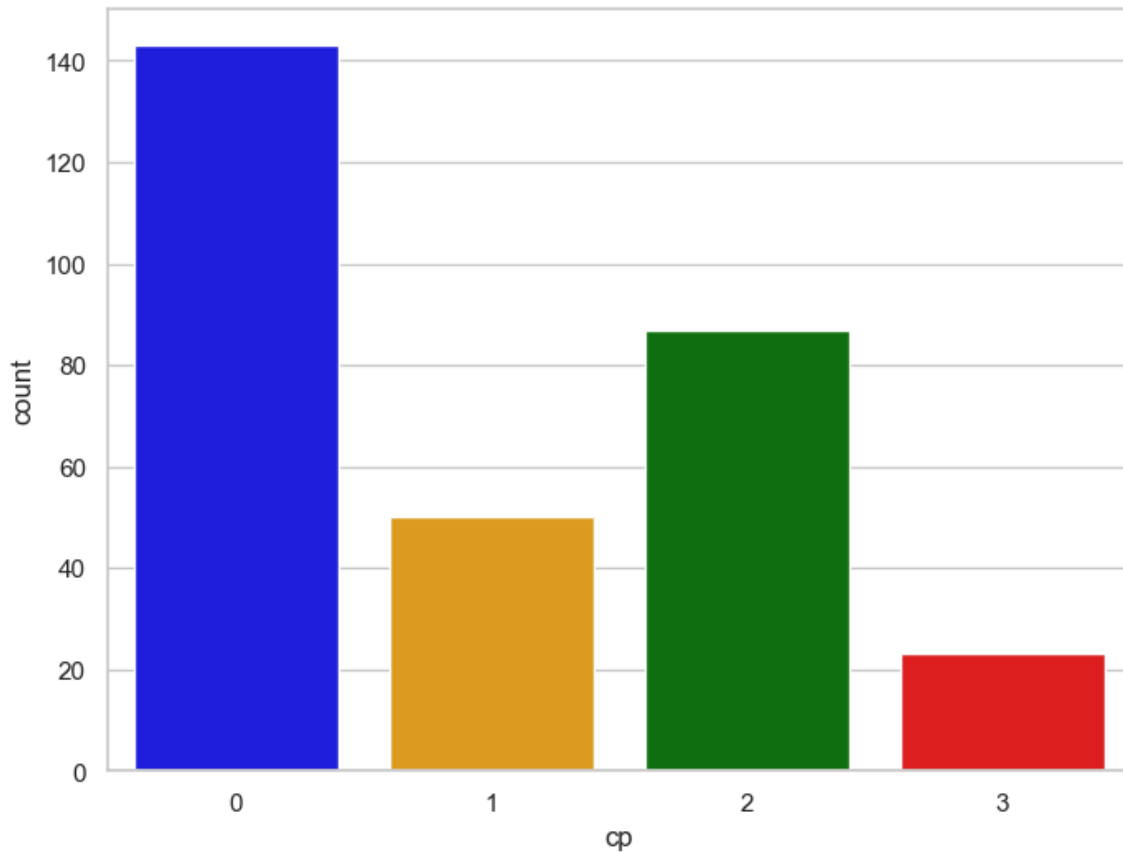
```
4
```

```
df['cp'].value_counts()
```



```
cp
0    143
2     87
1     50
3     23
Name: count, dtype: int64
```

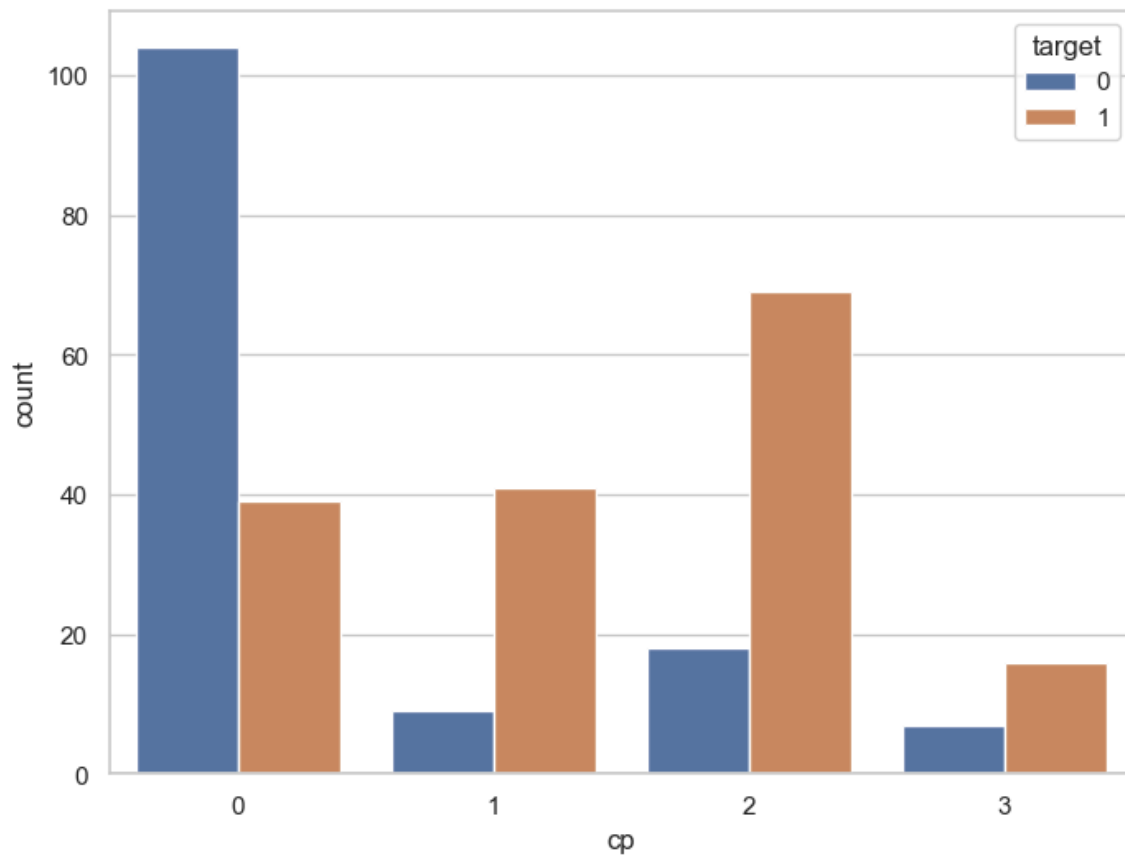
```
f,ax=plt.subplots(figsize=(8,6))
ax=sns.countplot(x='cp',data=df,palette=['blue','orange','green','red'])
plt.show()
```



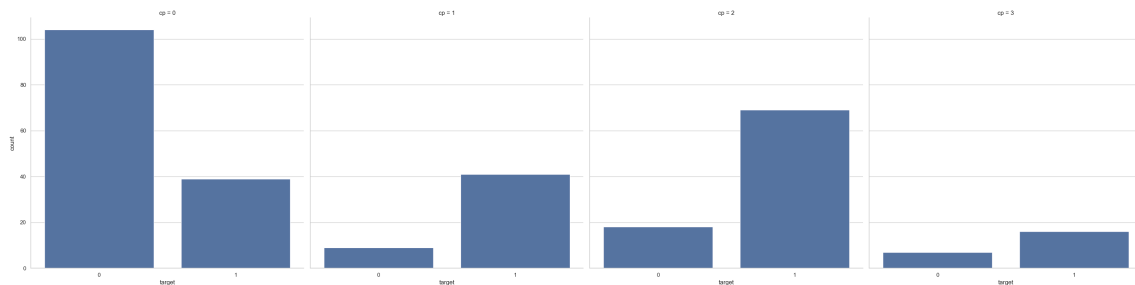
```
df.groupby('cp')['target'].value_counts()
```

```
cp target
0  0    104
   1     39
1  1     41
   0      9
2  1     69
   0     18
3  1     16
   0      7
Name: count, dtype: int64
```

```
f, ax = plt.subplots(figsize=(8, 6))
ax = sns.countplot(x="cp", hue="target", data=df)
plt.show()
```



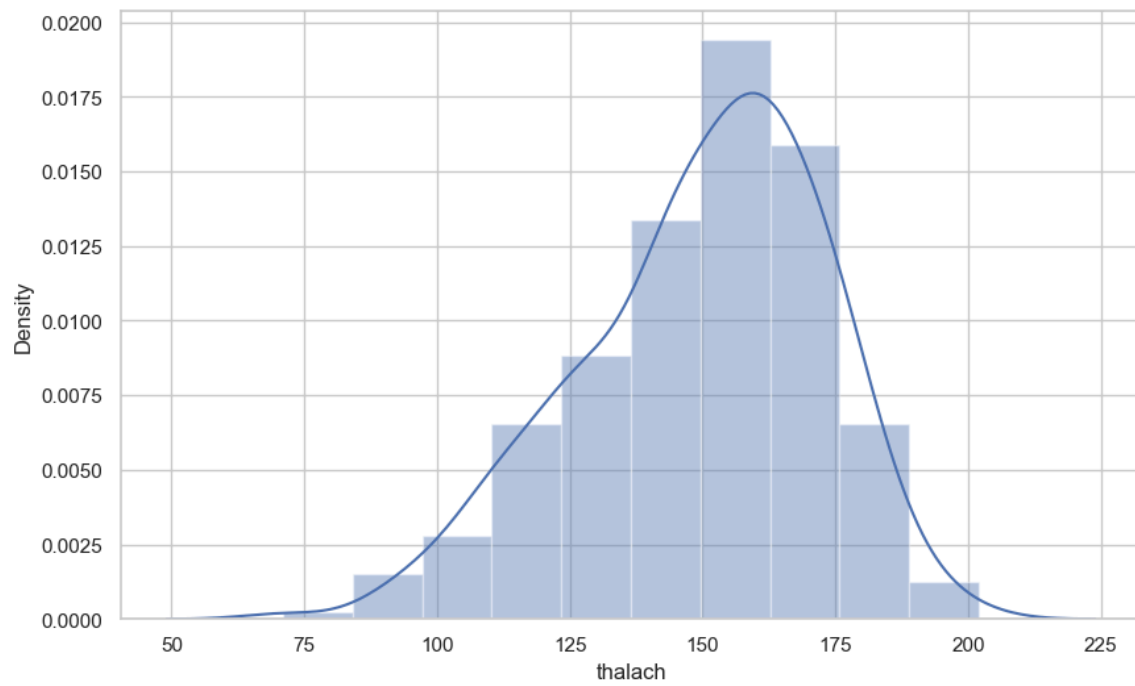
```
ax = sns.catplot(x="target", col="cp", data=df, kind="count", height=8,
                 aspect=1)
```



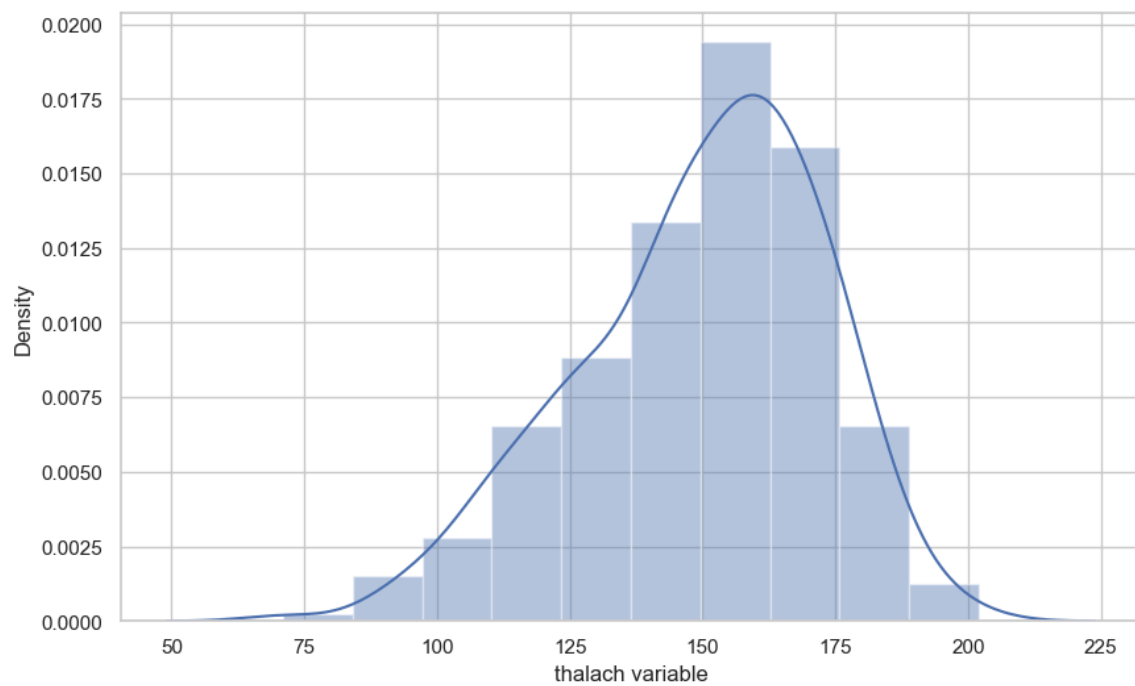
```
df['thalach'].nunique()
```

91

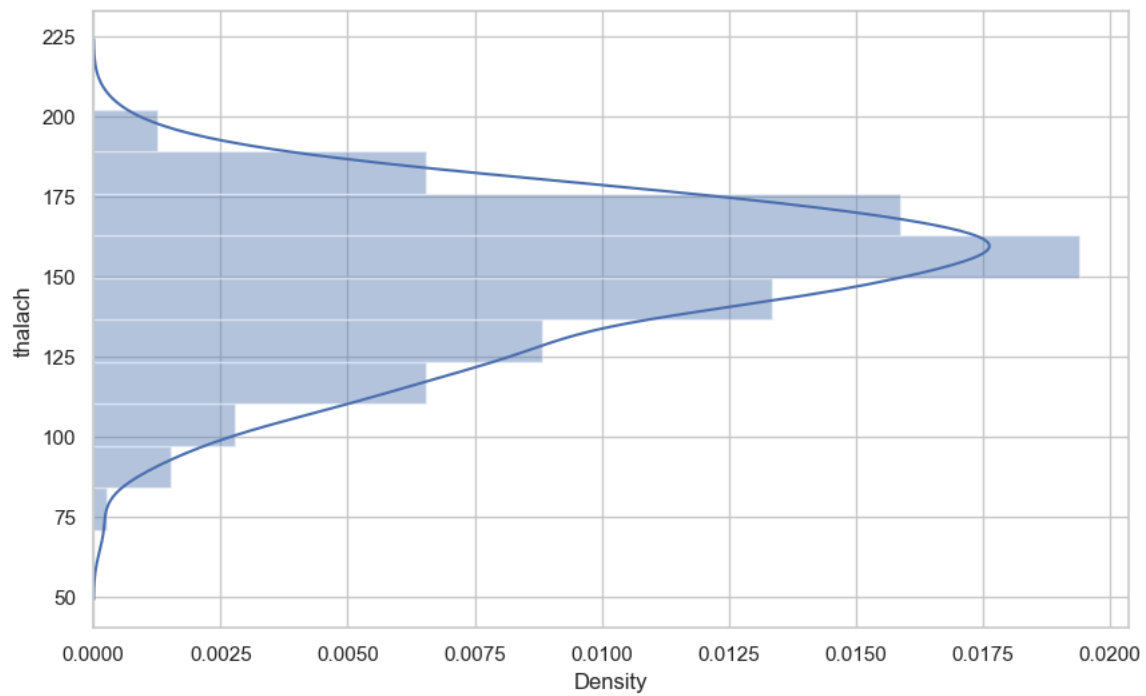
```
f, ax = plt.subplots(figsize=(10,6))
x = df['thalach']
ax = sns.distplot(x, bins=10)
plt.show()
```



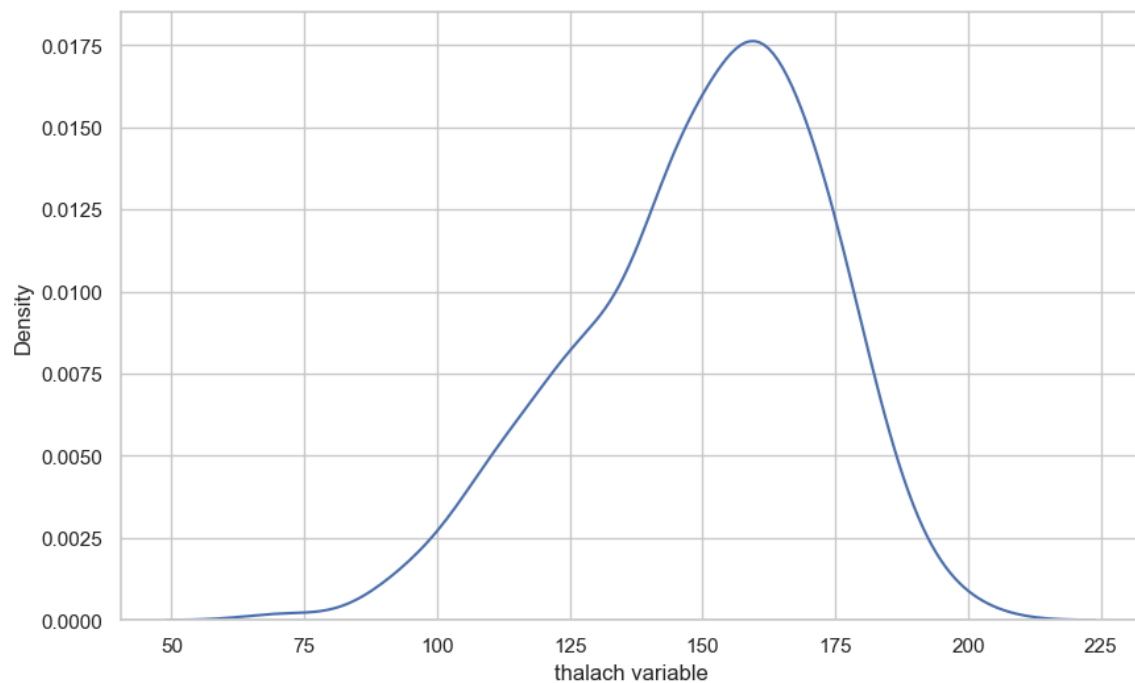
```
f, ax = plt.subplots(figsize=(10,6))
x = df['thalach']
x = pd.Series(x, name="thalach variable")
ax = sns.distplot(x, bins=10)
plt.show()
```



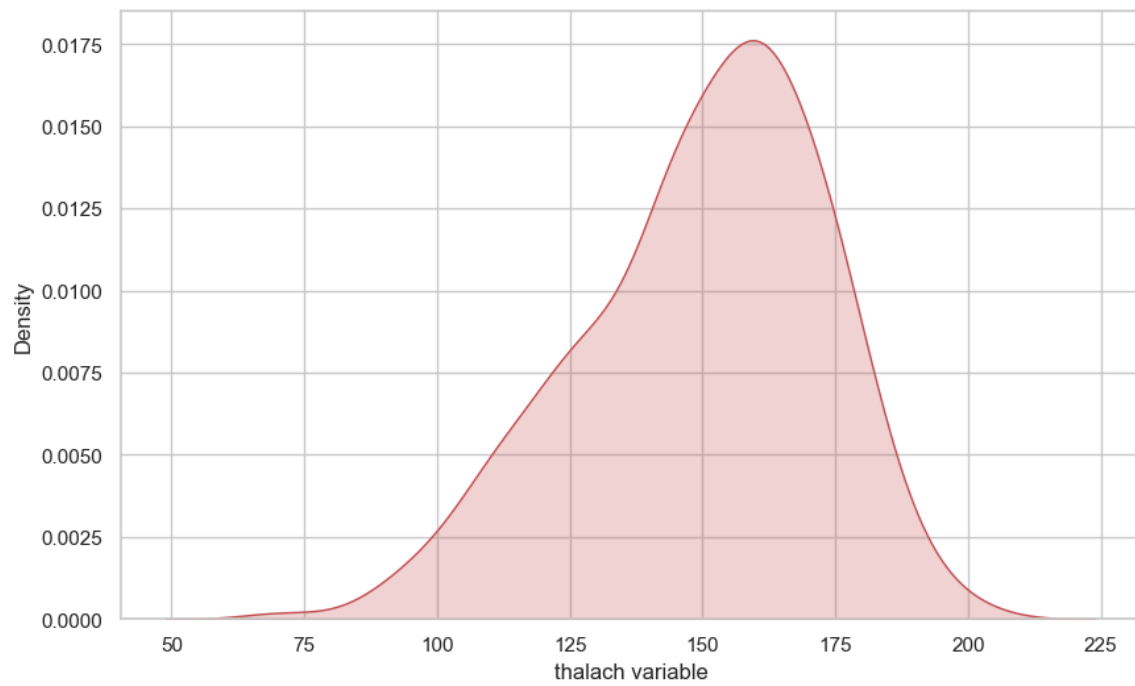
```
f, ax = plt.subplots(figsize=(10,6))
x = df['thalach']
ax = sns.distplot(x, bins=10, vertical=True)
plt.show()
```



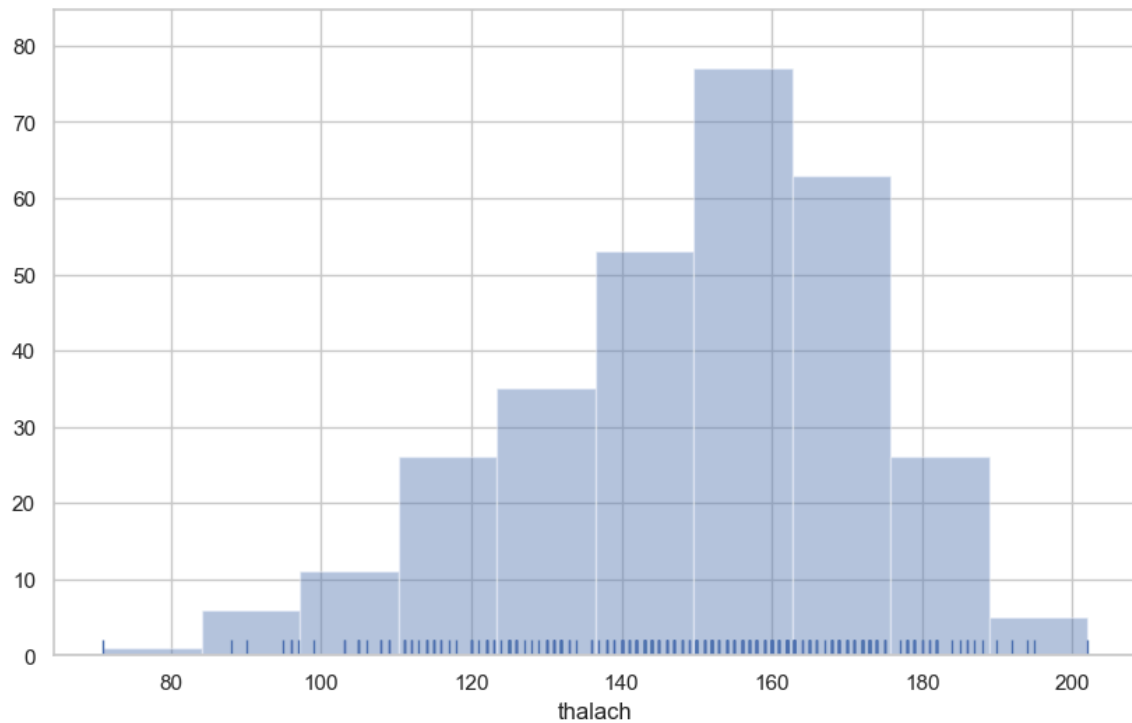
```
f, ax = plt.subplots(figsize=(10,6))
x = df['thalach']
x = pd.Series(x, name="thalach variable")
ax = sns.kdeplot(x)
plt.show()
```



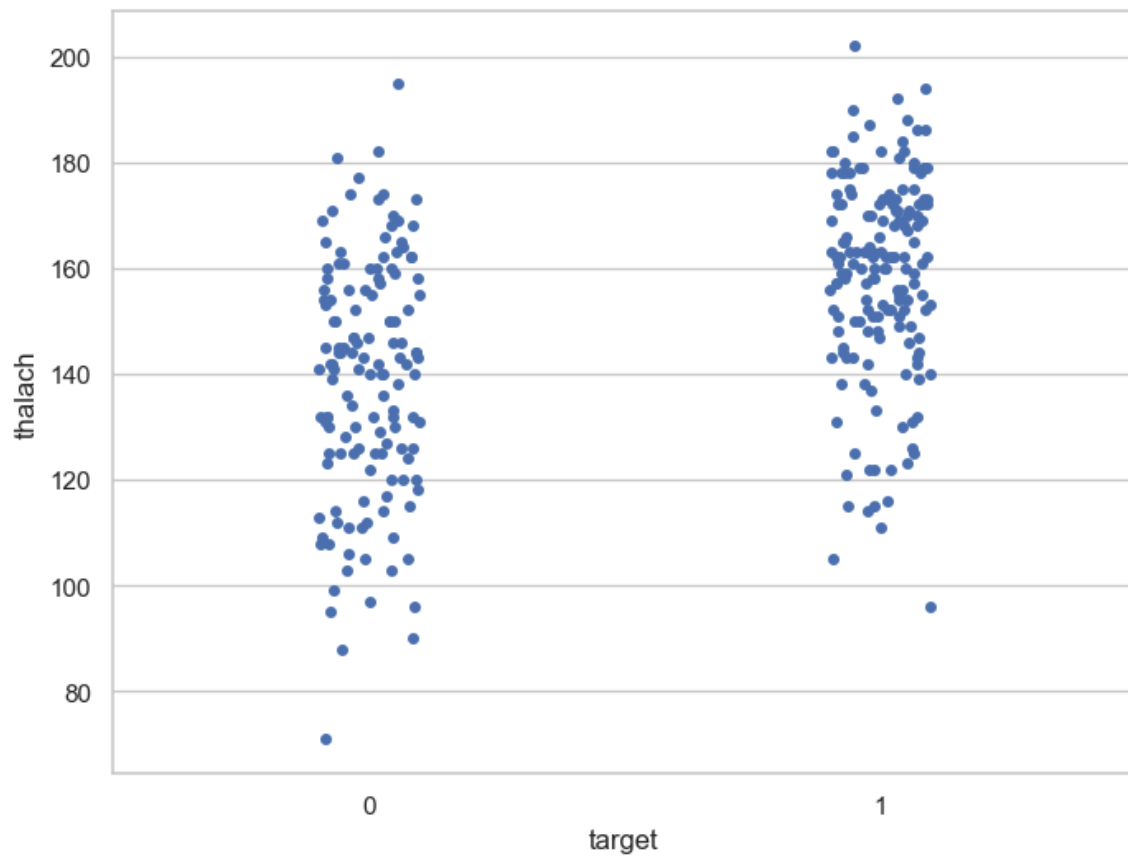
```
f, ax = plt.subplots(figsize=(10,6))
x = df['thalach']
x = pd.Series(x, name="thalach variable")
ax = sns.kdeplot(x, shade=True, color='r')
plt.show()
```



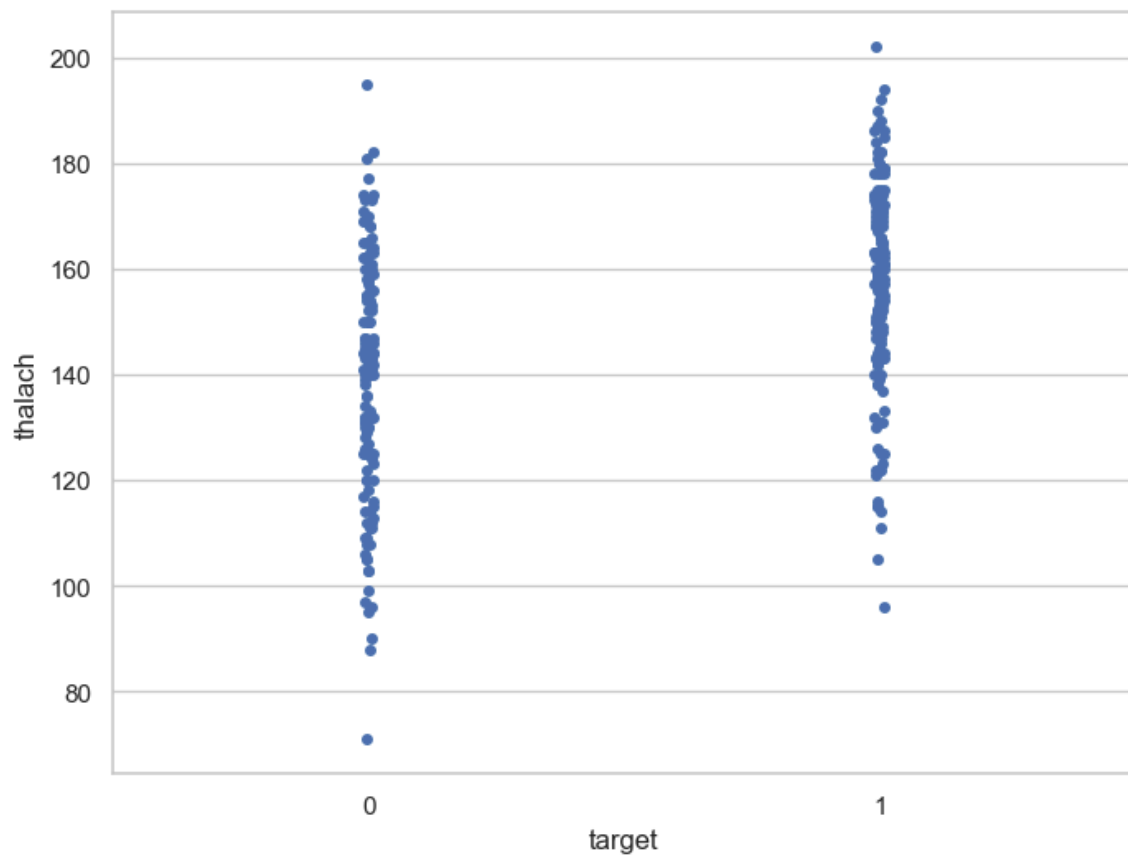
```
f, ax = plt.subplots(figsize=(10,6))
x = df['thalach']
ax = sns.distplot(x, kde=False, rug=True, bins=10)
plt.show()
```



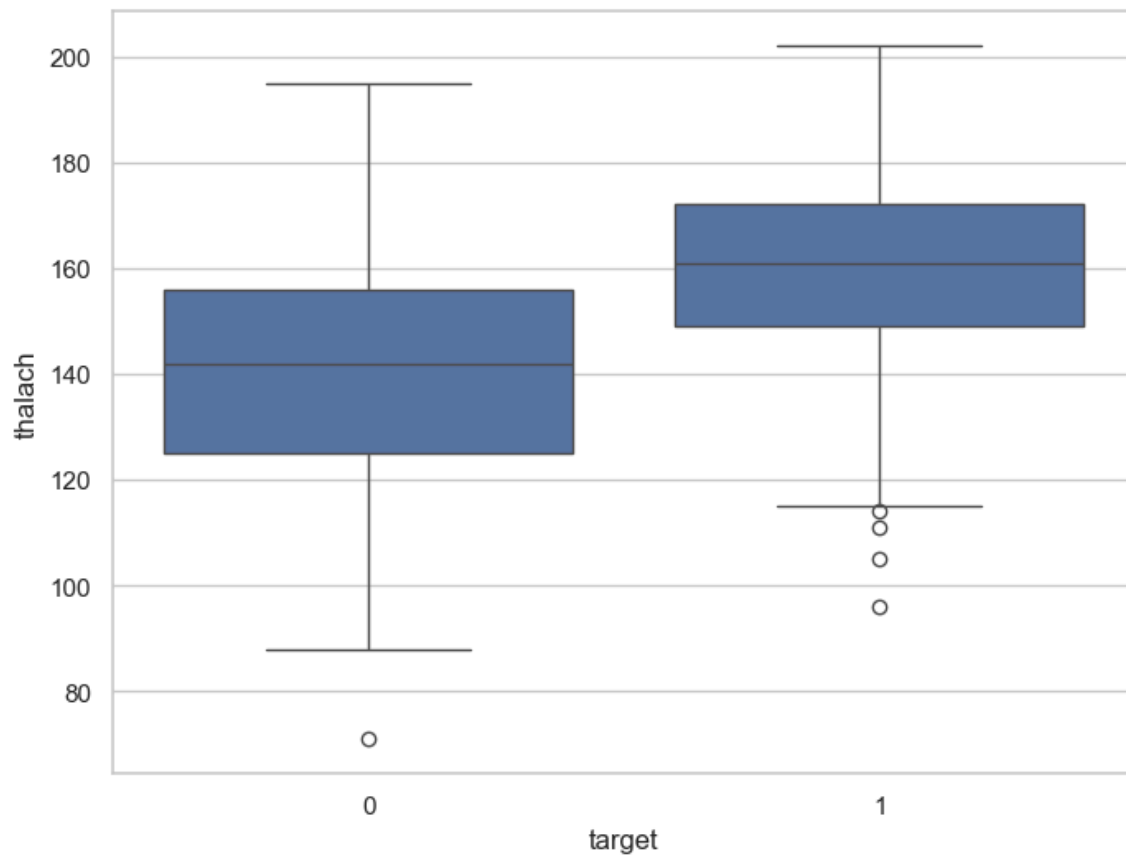
```
f, ax = plt.subplots(figsize=(8, 6))
sns.stripplot(x="target", y="thalach", data=df)
plt.show()
```



```
f, ax = plt.subplots(figsize=(8, 6))  
sns.stripplot(x="target", y="thalach", data=df, jitter = 0.01)  
plt.show()
```



```
f, ax = plt.subplots(figsize=(8, 6))
sns.boxplot(x="target", y="thalach", data=df)
plt.show()
```

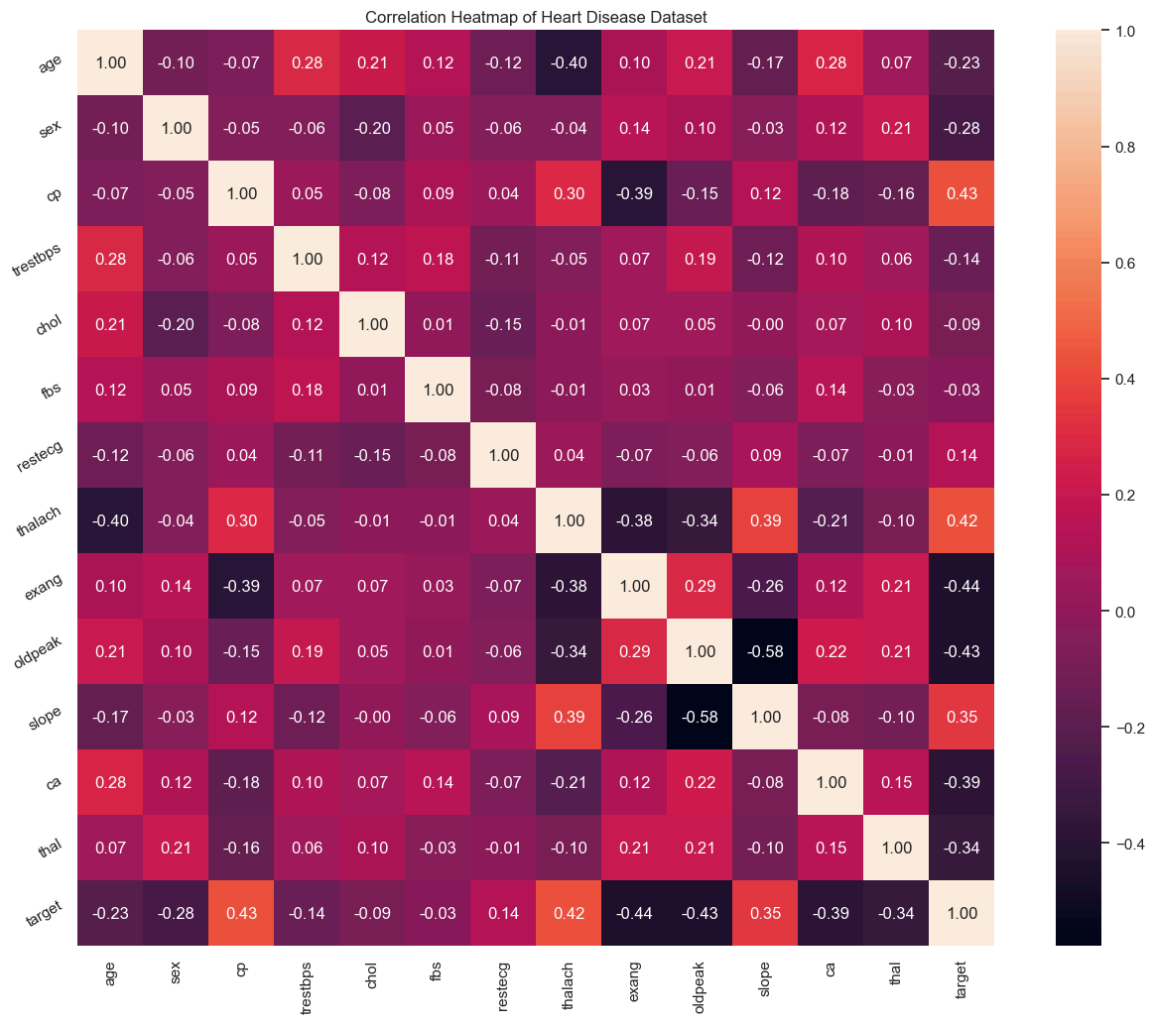


```
plt.figure(figsize=(16,12))
```

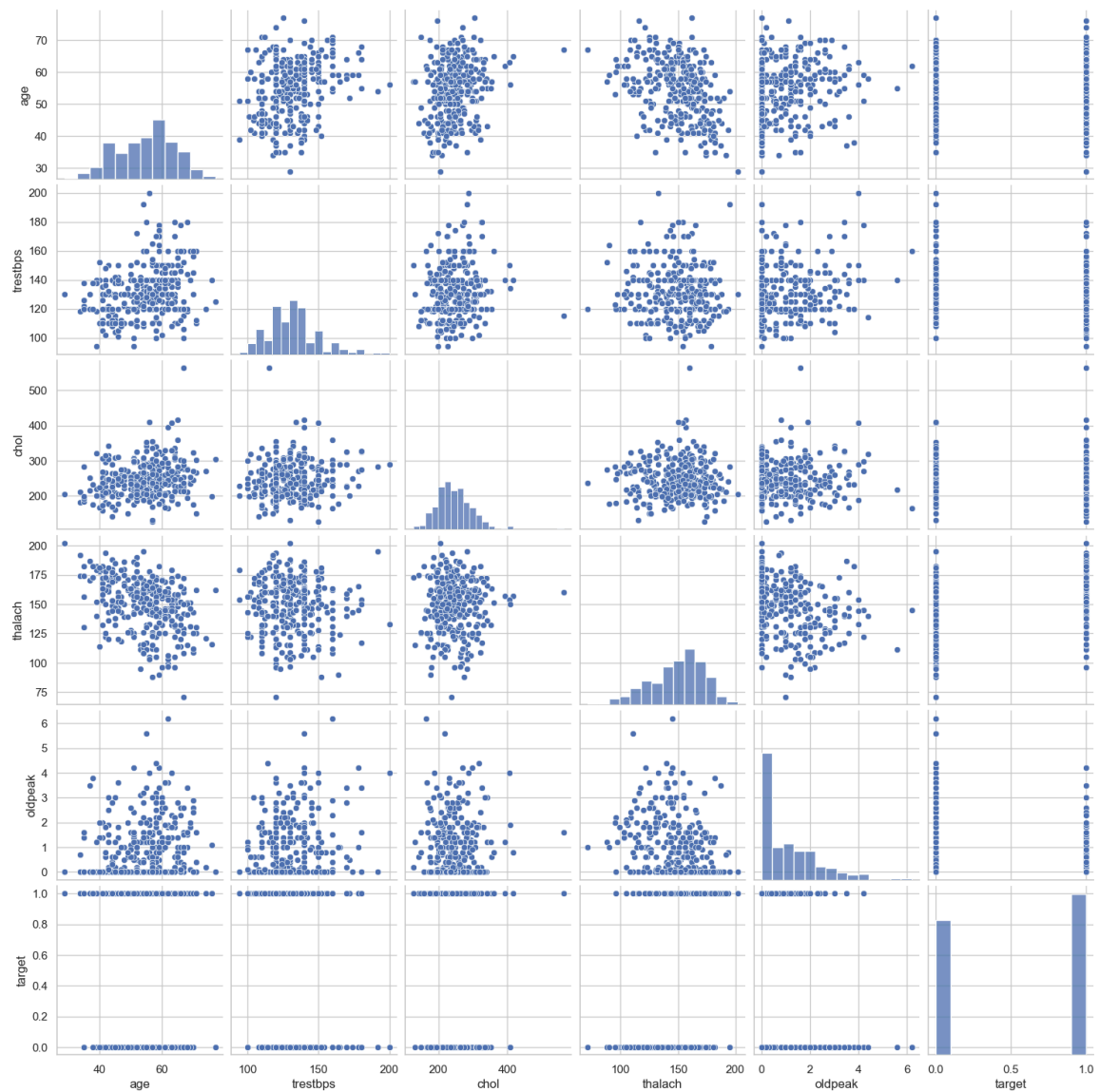
<Figure size 1600x1200 with 0 Axes>

<Figure size 1600x1200 with 0 Axes>

```
plt.figure(figsize=(16,12))
plt.title('Correlation Heatmap of Heart Disease Dataset')
a = sns.heatmap(correlation, square=True, annot=True, fmt='.2f',
                 linecolor='white')
a.set_xticklabels(a.get_xticklabels(), rotation=90)
a.set_yticklabels(a.get_yticklabels(), rotation=30)
plt.show()
```



```
num_var = ['age', 'trestbps', 'chol', 'thalach', 'oldpeak', 'target' ]
sns.pairplot(df[num_var], kind='scatter', diag_kind='hist')
plt.show()
```

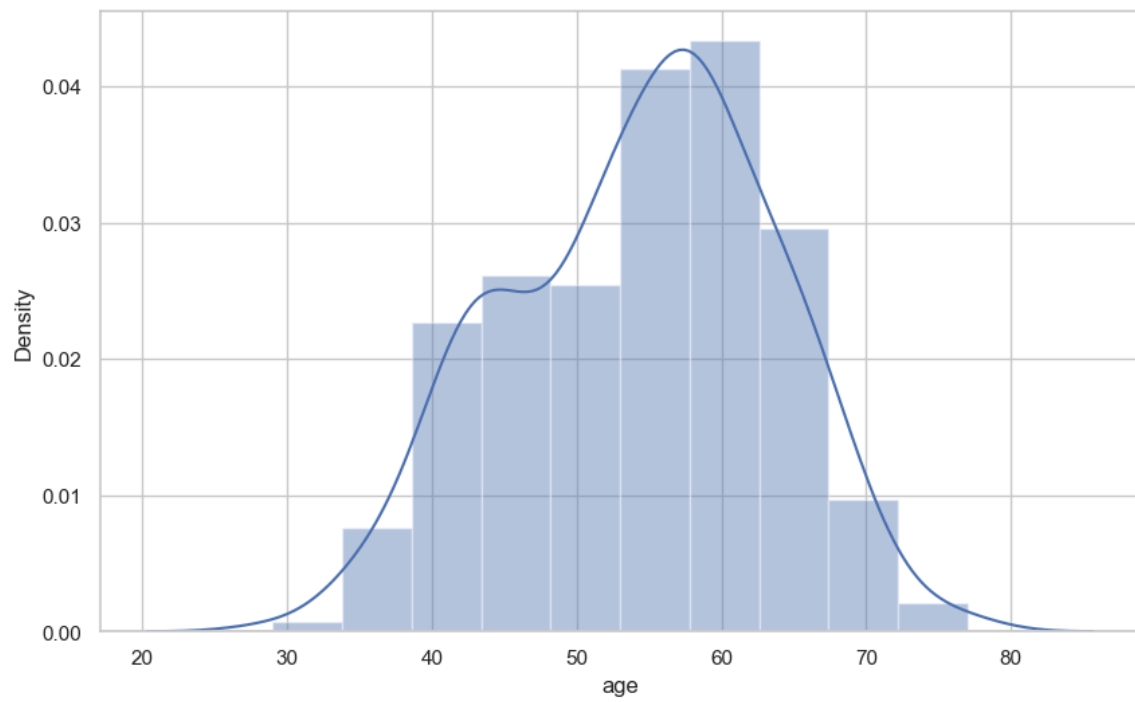
```
df['age'].nunique()
```

```
41
```

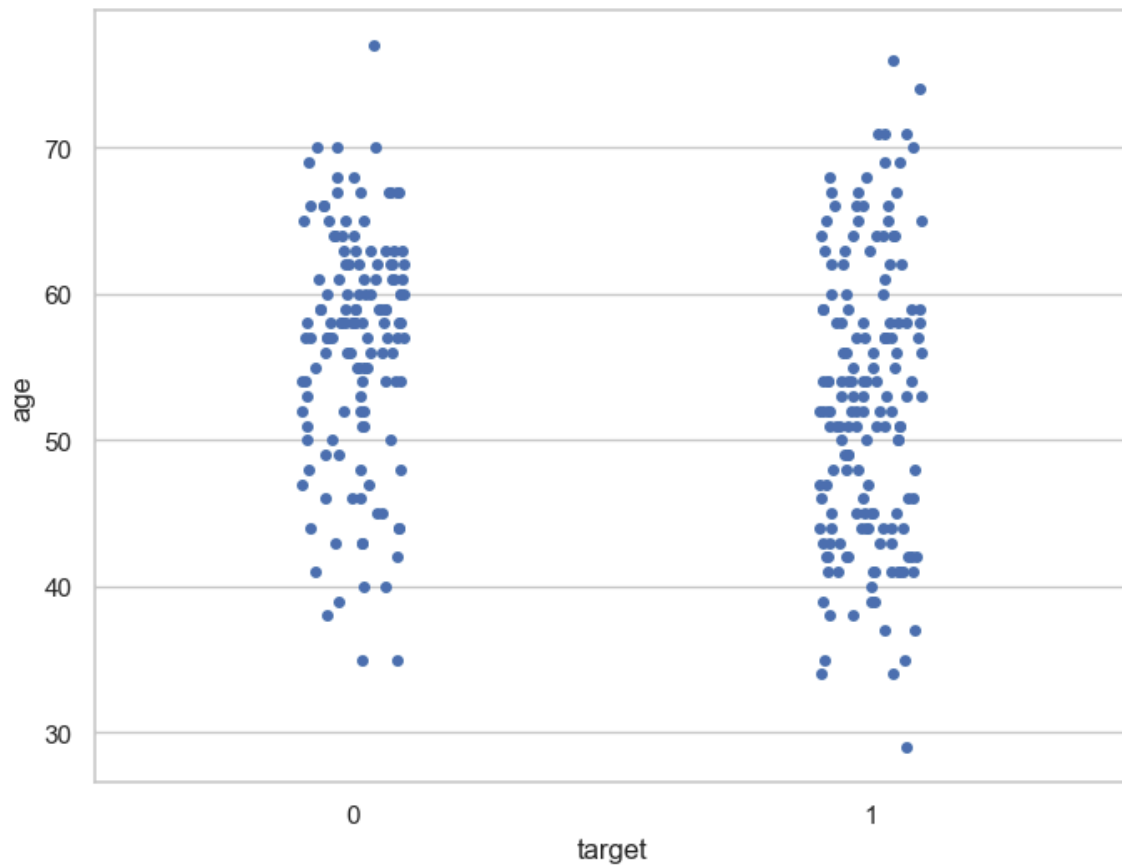
```
df['age'].describe()
```

```
count    303.000000
mean      54.366337
std       9.082101
min       29.000000
25%      47.500000
50%      55.000000
75%      61.000000
max       77.000000
Name: age, dtype: float64
```

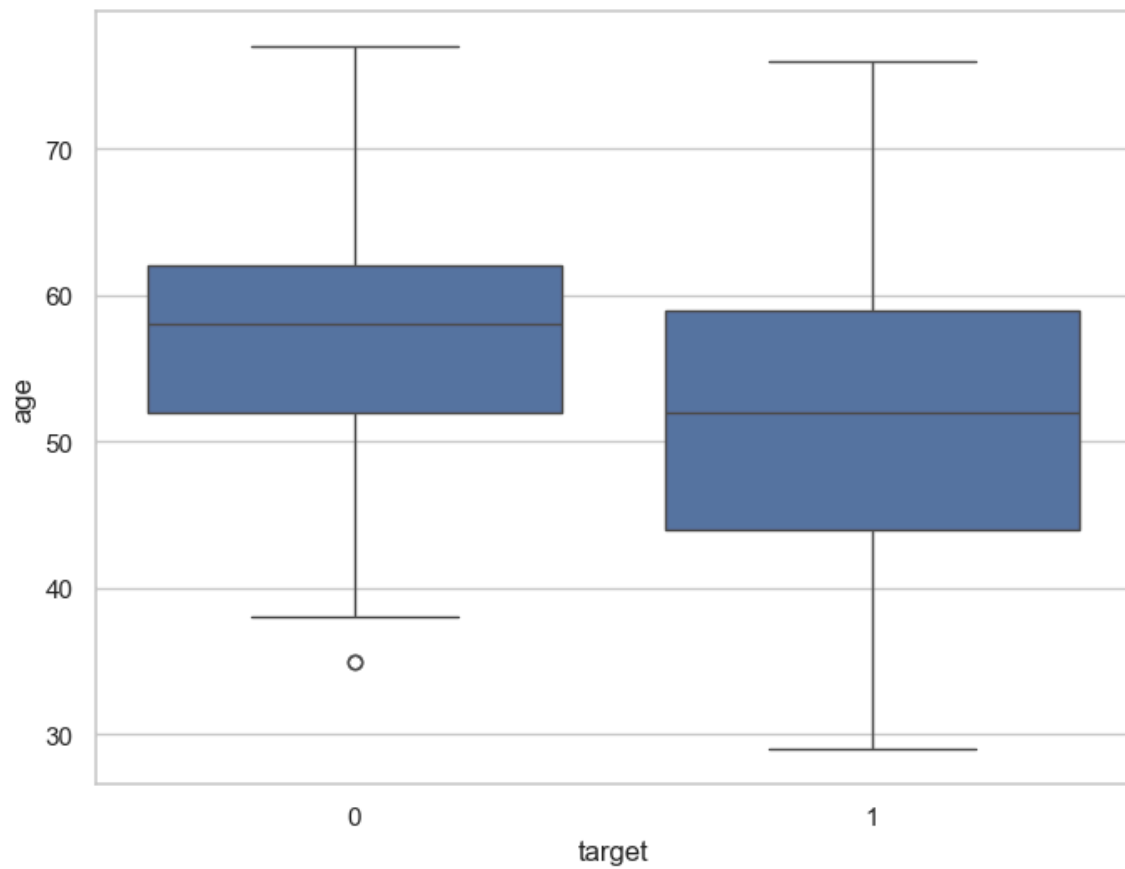
```
f, ax = plt.subplots(figsize=(10,6))
x = df['age']
ax = sns.distplot(x, bins=10)
plt.show()
```



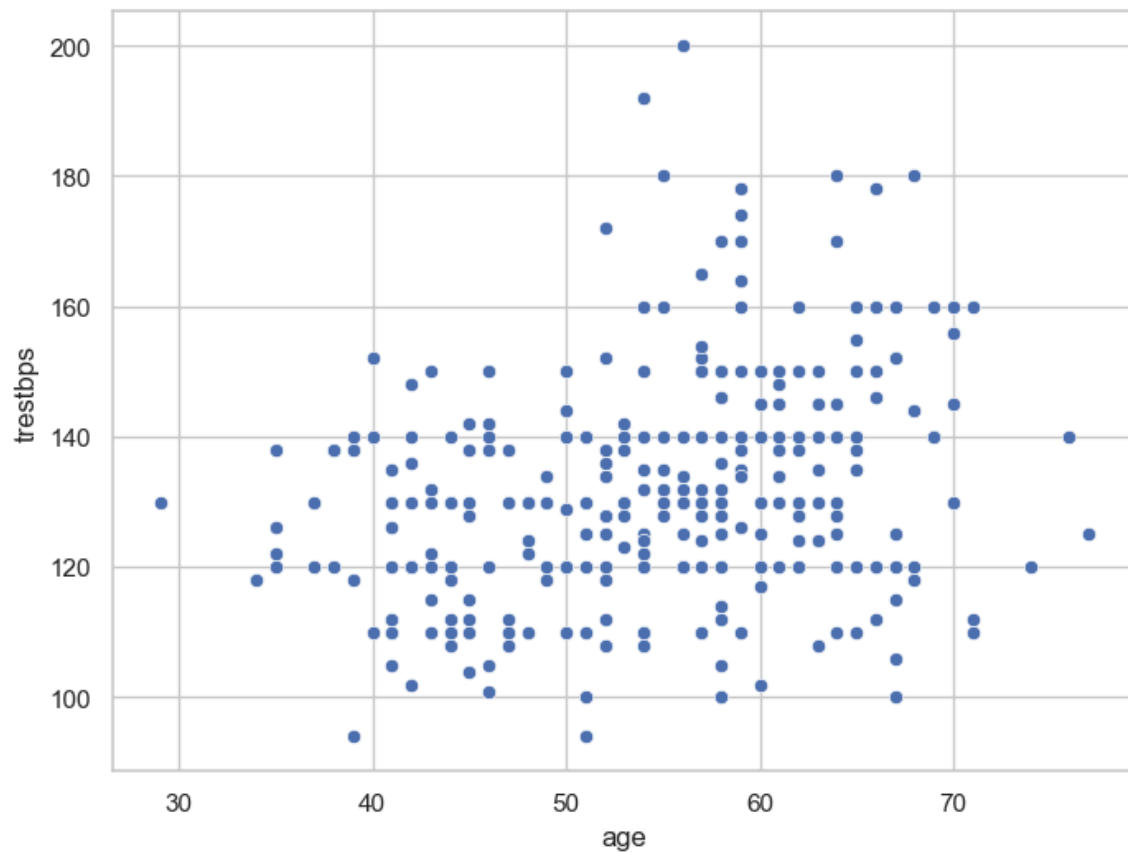
```
f, ax = plt.subplots(figsize=(8, 6))
sns.stripplot(x="target", y="age", data=df)
plt.show()
```



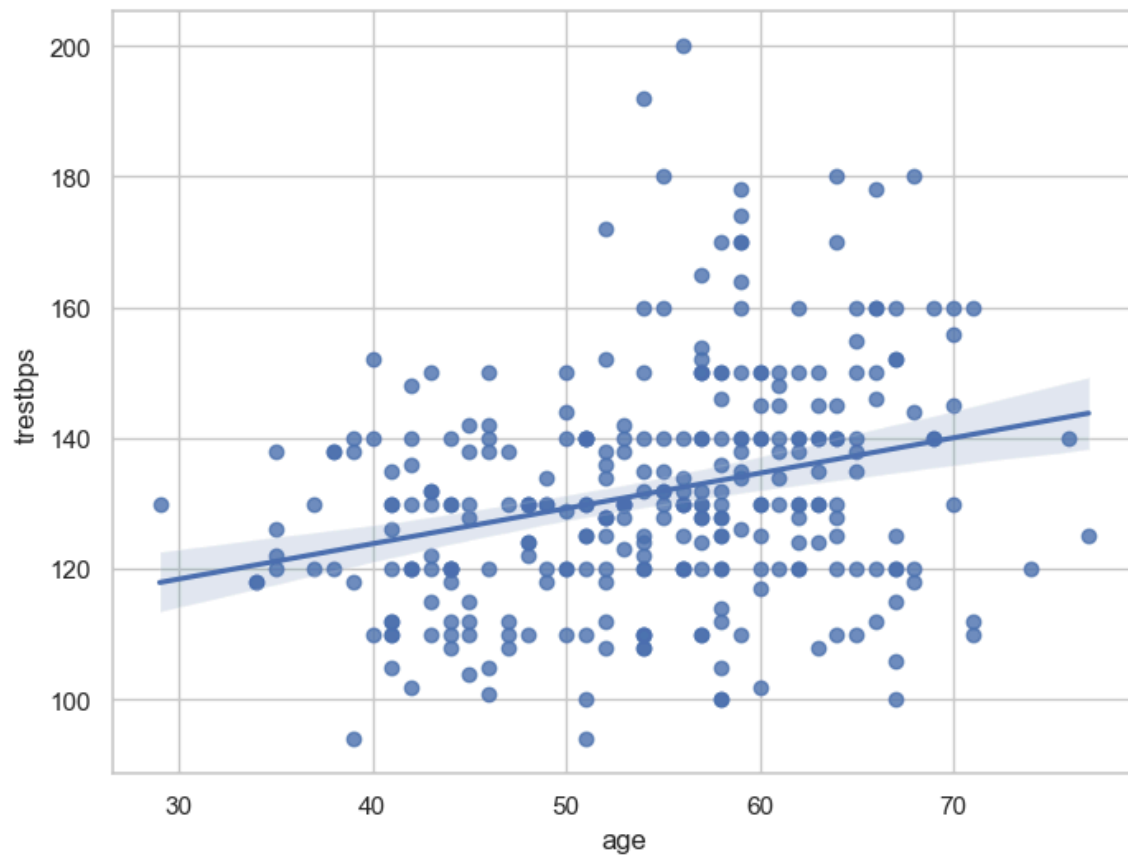
```
f, ax = plt.subplots(figsize=(8, 6))
sns.boxplot(x="target", y="age", data=df)
plt.show()
```



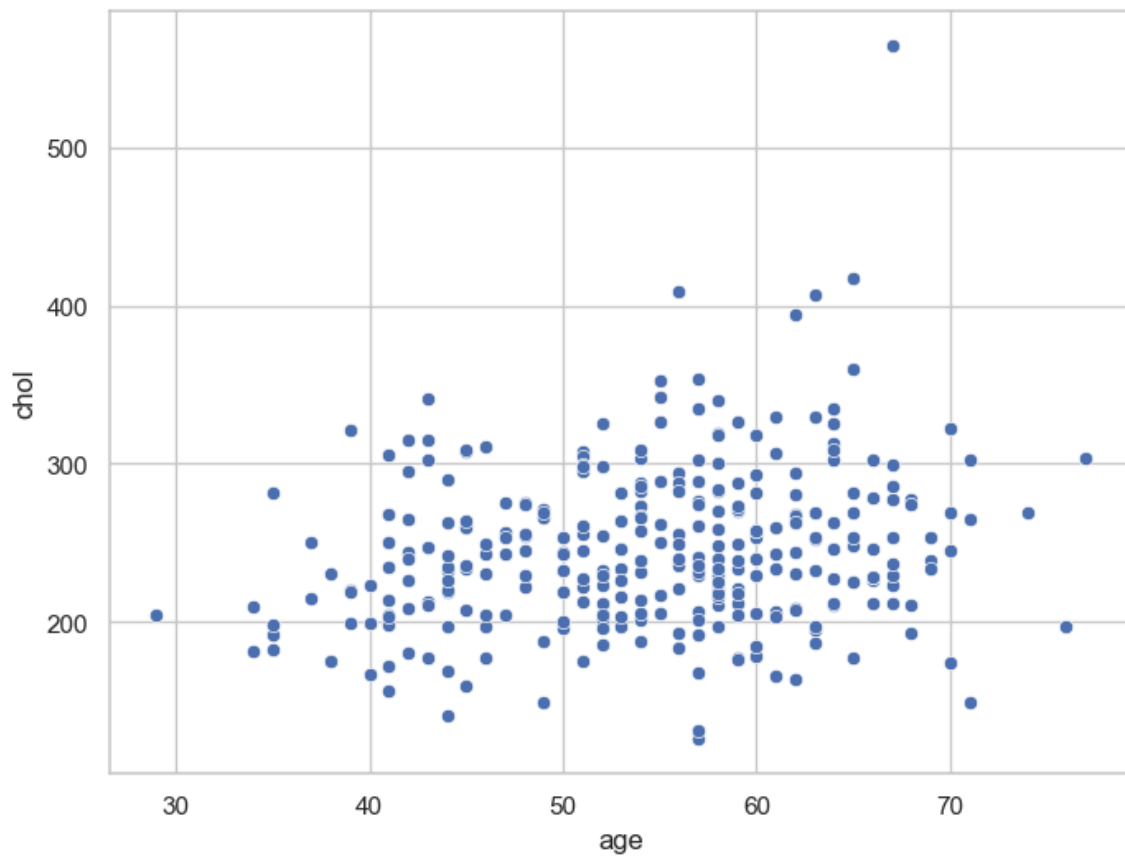
```
f, ax = plt.subplots(figsize=(8, 6))  
ax = sns.scatterplot(x="age", y="trestbps", data=df)  
plt.show()
```



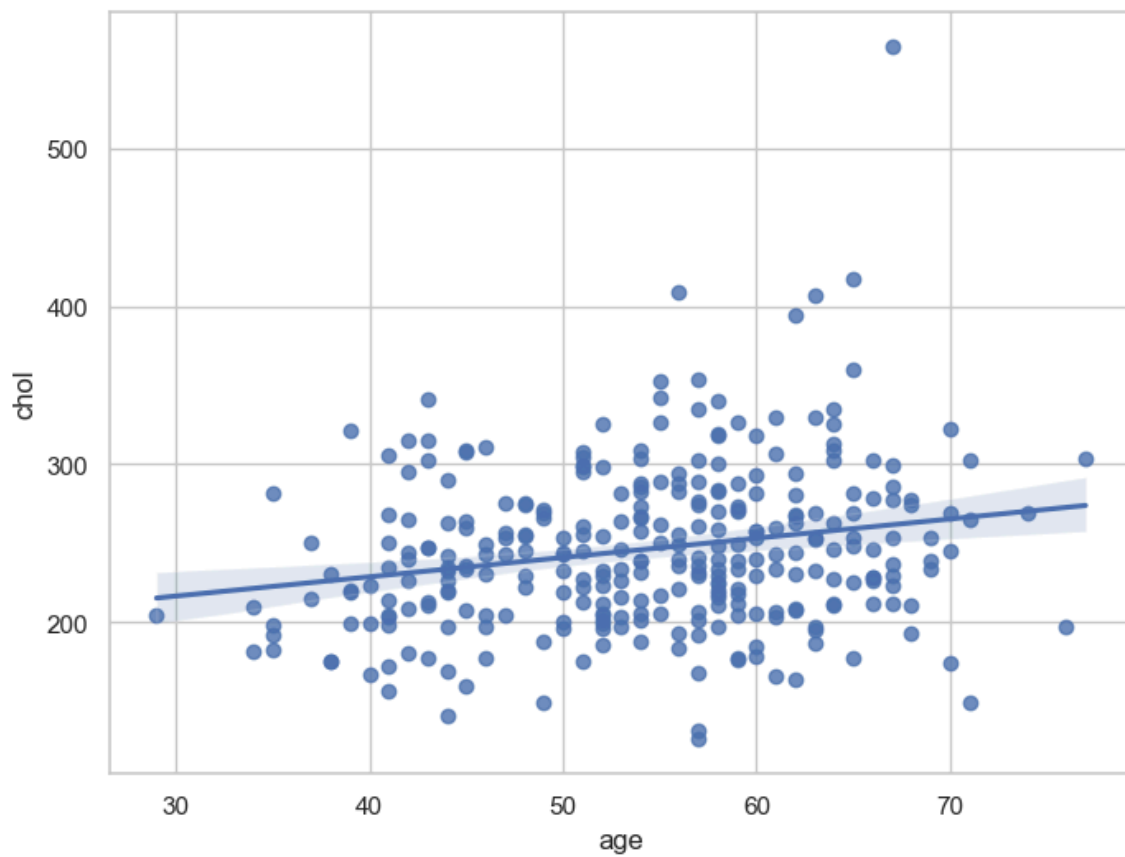
```
f, ax = plt.subplots(figsize=(8, 6))
ax = sns.regplot(x="age", y="trestbps", data=df)
plt.show()
```



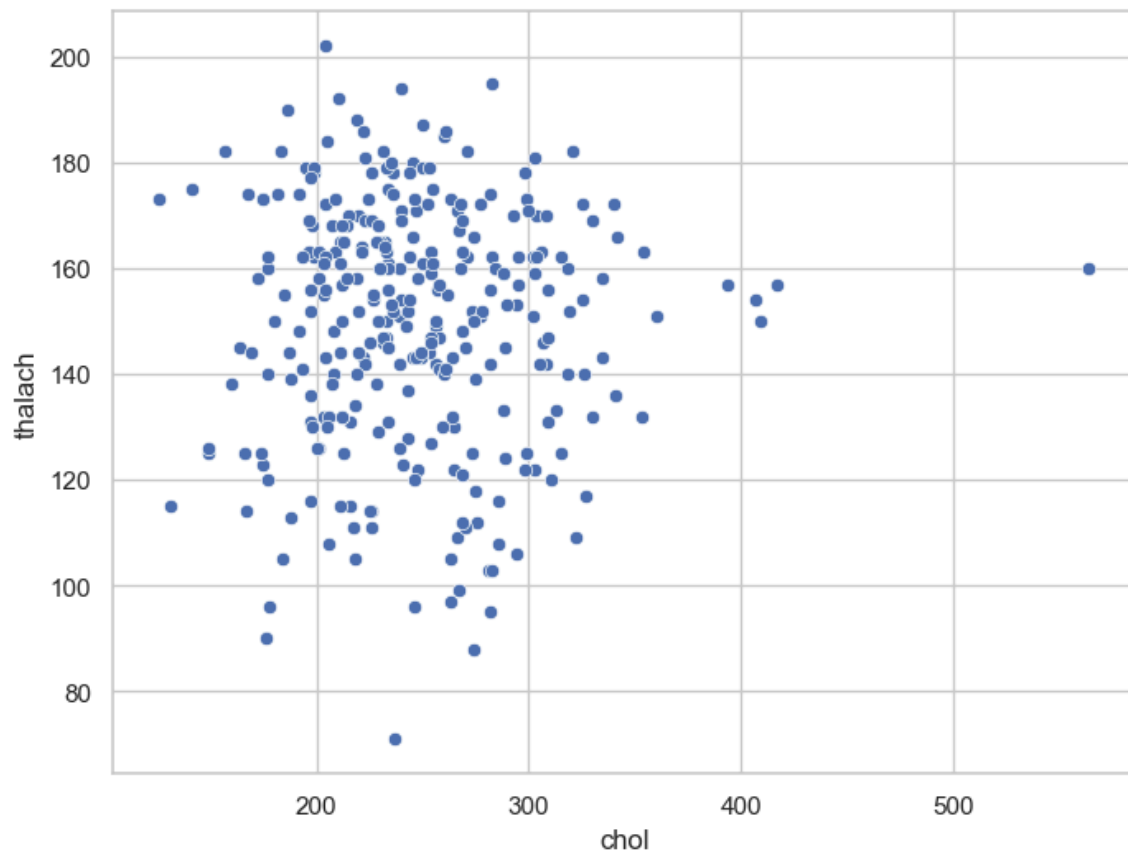
```
f, ax = plt.subplots(figsize=(8, 6))
ax = sns.scatterplot(x="age", y="chol", data=df)
plt.show()
```



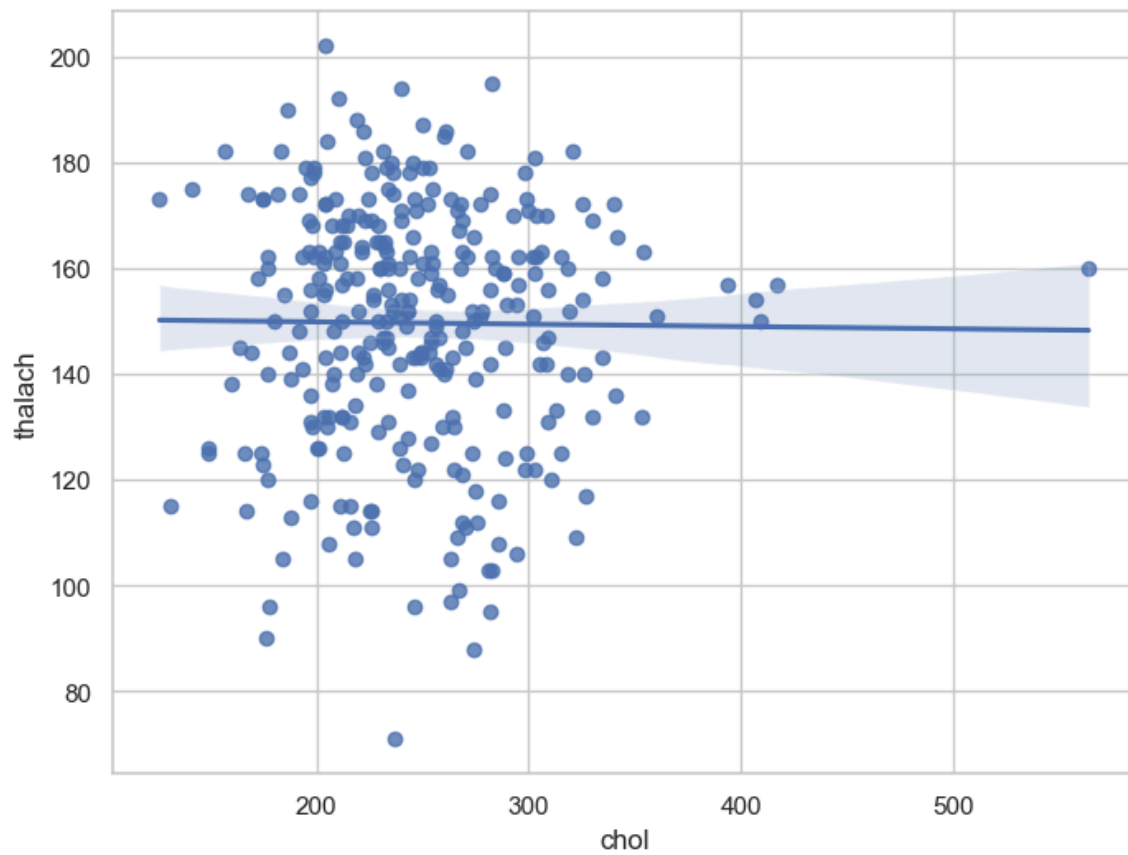
```
f, ax = plt.subplots(figsize=(8, 6))  
ax = sns.regplot(x="age", y="chol", data=df)  
plt.show()
```



```
f, ax = plt.subplots(figsize=(8, 6))
ax = sns.scatterplot(x="chol", y="thalach", data=df)
plt.show()
```



```
f, ax = plt.subplots(figsize=(8, 6))
ax = sns.regplot(x="chol", y="thalach", data=df)
plt.show()
```



check for missing values

```
df.isnull().sum()
```

```
age          0
sex          0
cp           0
trestbps     0
chol         0
fbs          0
restecg      0
thalach      0
exang        0
oldpeak      0
slope        0
ca           0
thal         0
target       0
dtype: int64
```

```
assert pd.notnull(df).all().all()
```

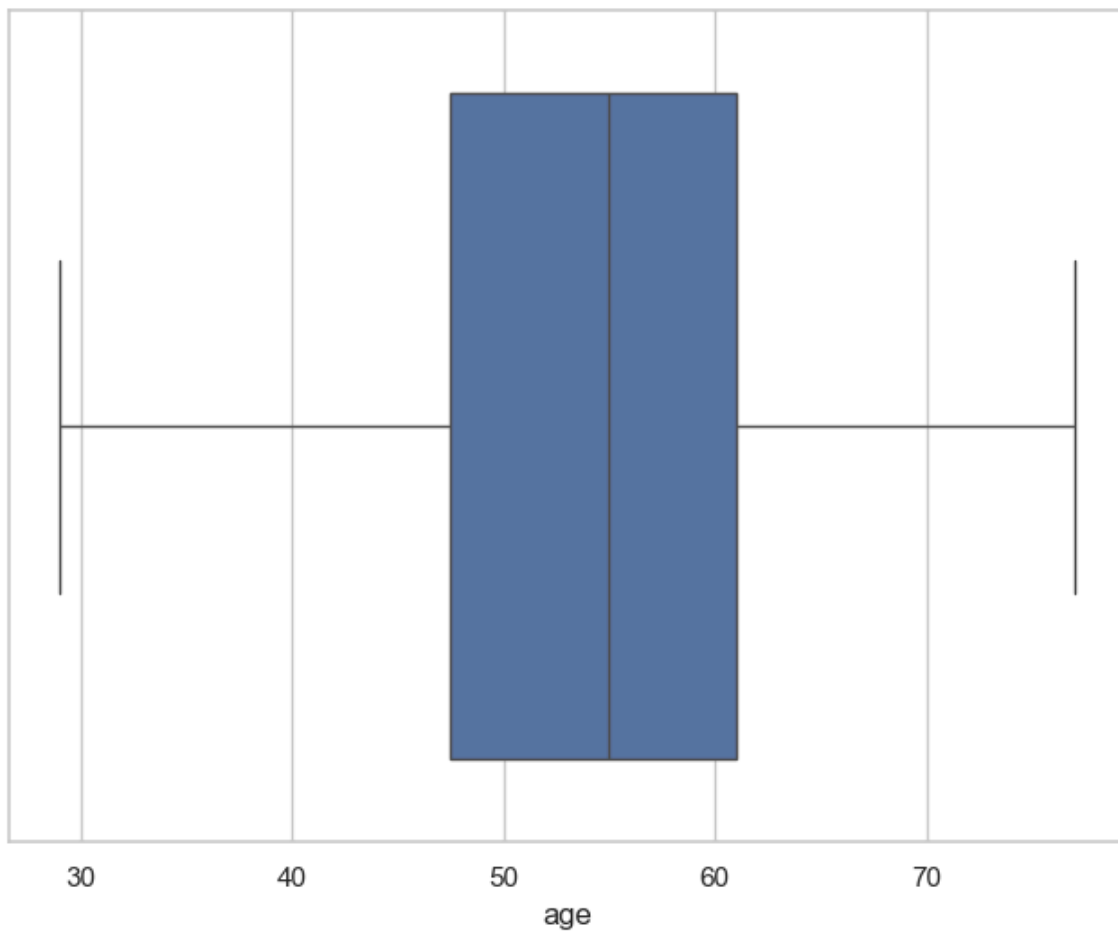
#assert all values are greater than or equal to 0

```
assert (df >= 0).all().all()
```

```
df['age'].describe()
```

```
count    303.000000
mean      54.366337
std       9.082101
min       29.000000
25%      47.500000
50%      55.000000
75%      61.000000
max       77.000000
Name: age, dtype: float64
```

```
f, ax = plt.subplots(figsize=(8, 6))
sns.boxplot(x=df["age"])
plt.show()
```

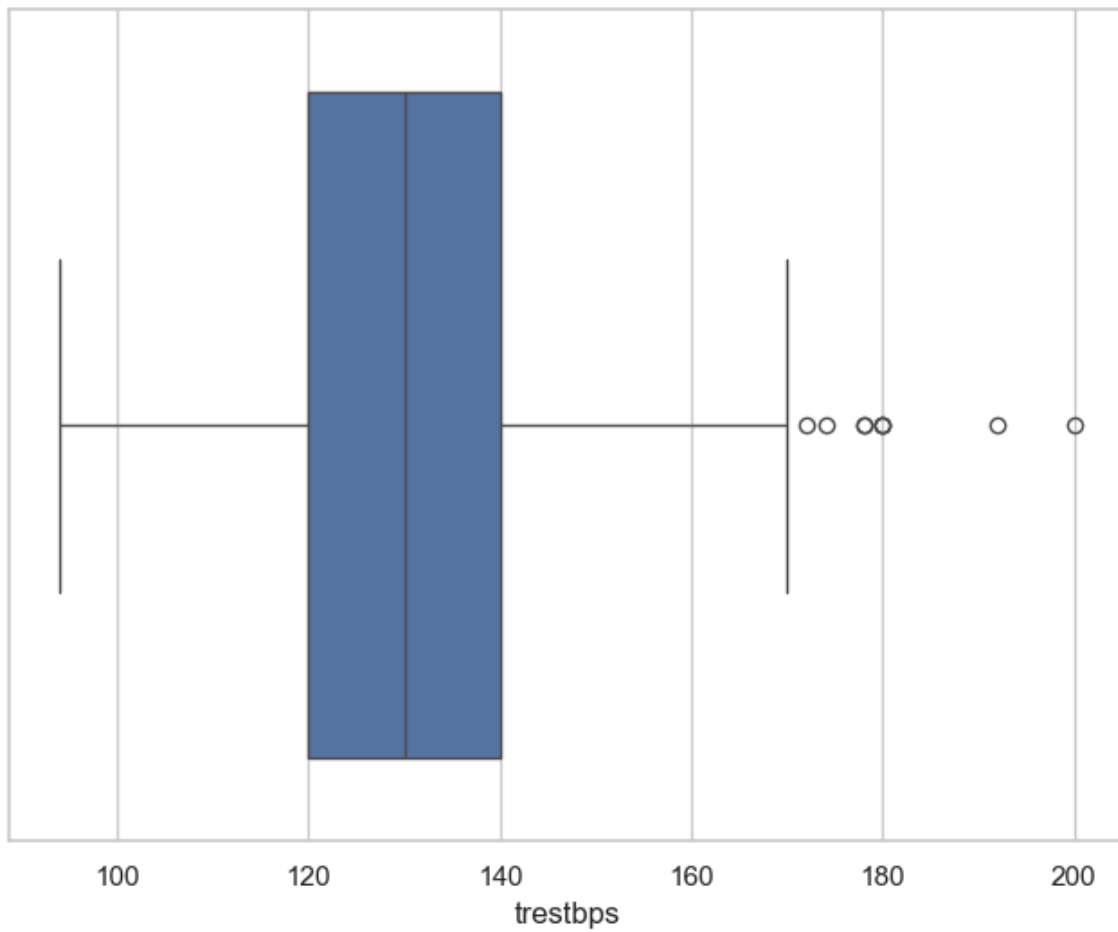


```
df['trestbps'].describe()
```

```
count    303.000000
mean     131.623762
std      17.538143
min      94.000000
25%     120.000000
50%     130.000000
75%     140.000000
max     200.000000
Name: trestbps, dtype: float64
```



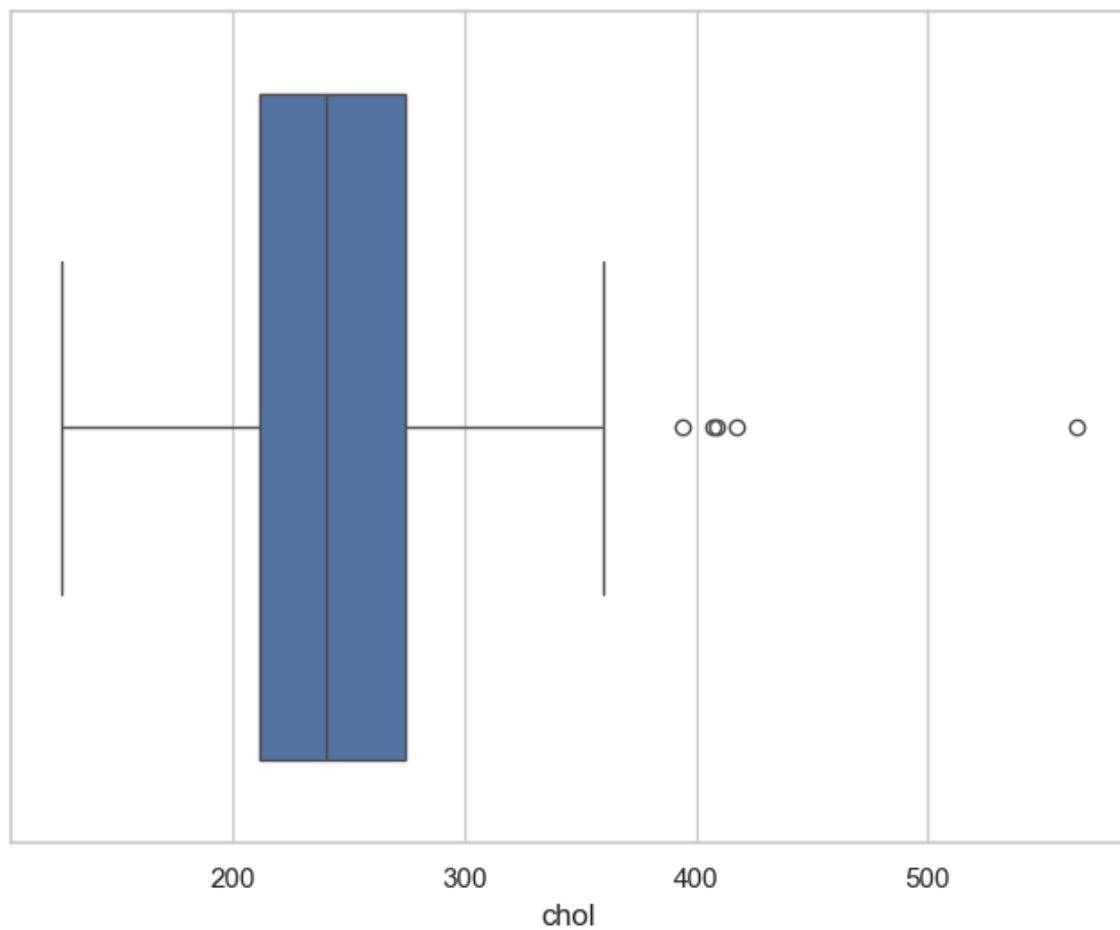
```
f, ax = plt.subplots(figsize=(8, 6))
sns.boxplot(x=df["trestbps"])
plt.show()
```



```
df['chol'].describe()
```

```
count    303.000000
mean     246.264026
std       51.830751
min      126.000000
25%      211.000000
50%      240.000000
75%      274.500000
max       564.000000
Name: chol, dtype: float64
```

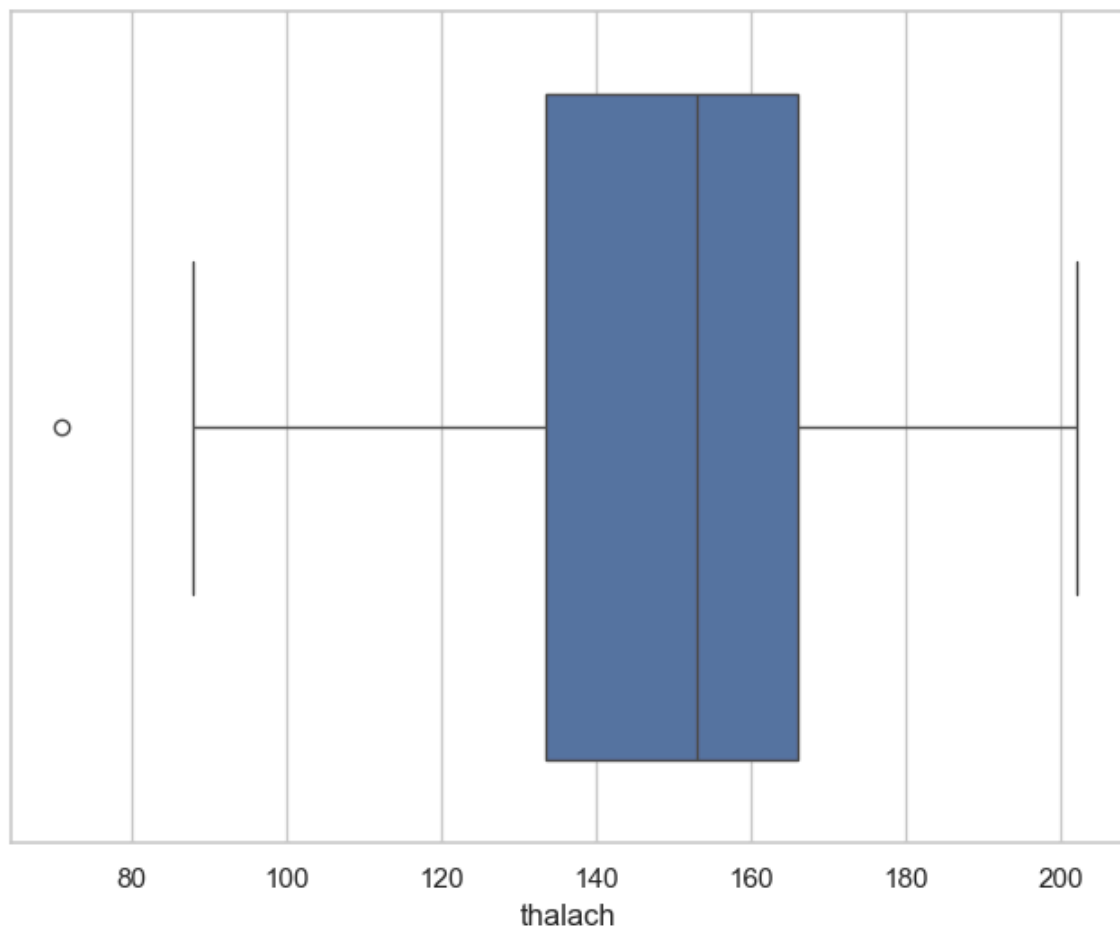
```
f, ax = plt.subplots(figsize=(8, 6))
sns.boxplot(x=df["chol"])
plt.show()
```



```
df['thalach'].describe()
```

```
count    303.000000
mean     149.646865
std       22.905161
min       71.000000
25%      133.500000
50%      153.000000
75%      166.000000
max       202.000000
Name: thalach, dtype: float64
```

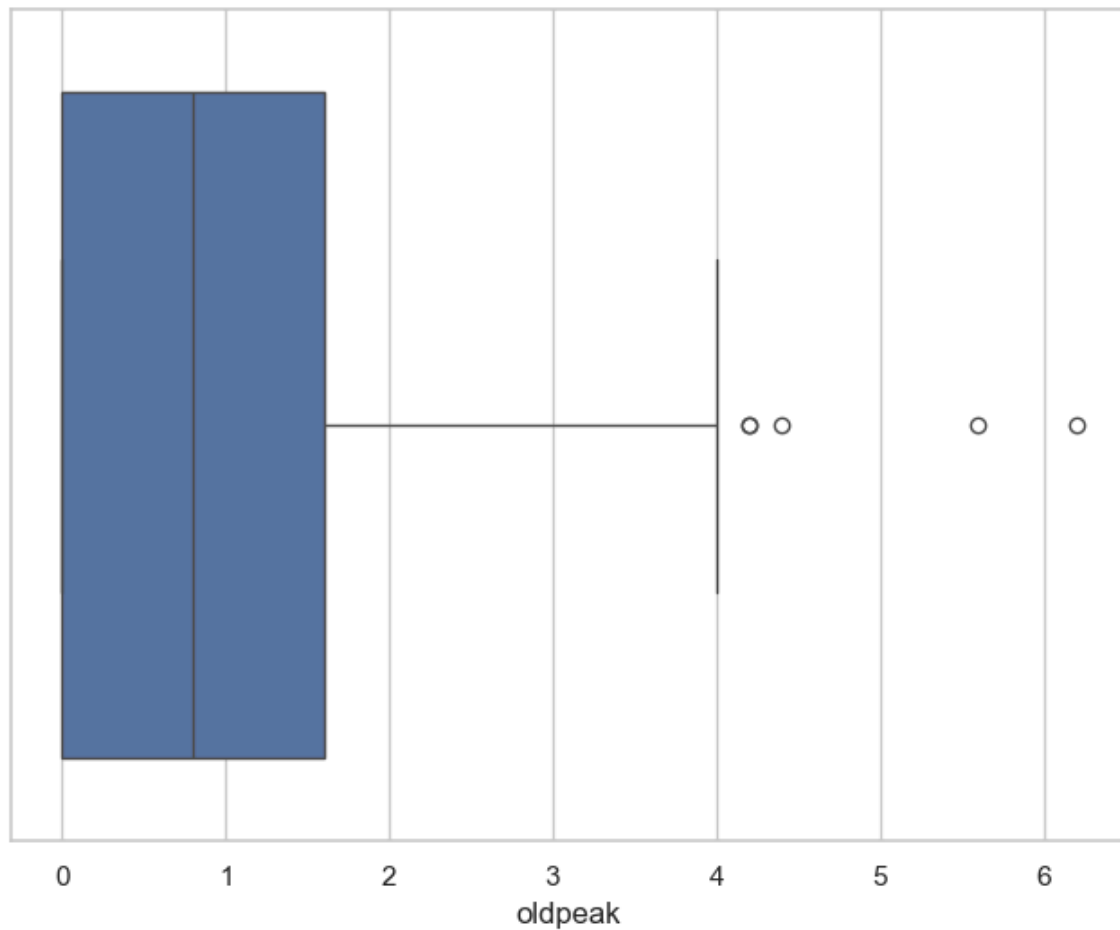
```
f, ax = plt.subplots(figsize=(8, 6))
sns.boxplot(x=df["thalach"])
plt.show()
```



```
df['oldpeak'].describe()
```

```
count    303.000000
mean       1.039604
std        1.161075
min         0.000000
25%         0.000000
50%         0.800000
75%         1.600000
max         6.200000
Name: oldpeak, dtype: float64
```

```
f, ax = plt.subplots(figsize=(8, 6))
sns.boxplot(x=df["oldpeak"])
plt.show()
```



```

Q1 = df["trestbps"].quantile(0.25)
Q3 = df["trestbps"].quantile(0.75)
IQR = Q3 - Q1
lower = Q1 - 1.5*IQR
upper = Q3 + 1.5*IQR
print("Lower Limit =", lower)
print("Upper Limit =", upper)
print(df[df["trestbps"] < lower].count())
print(df[df['trestbps']>upper].count())

```

```

Lower Limit = 90.0
Upper Limit = 170.0
age          0
sex          0
cp          0
trestbps     0
chol         0
fbs         0
restecg     0
thalach     0
exang       0
oldpeak     0
slope       0
ca          0
thal        0
target      0

```

```

dtype: int64
age      9
sex      9
cp       9
trestbps 9
chol     9
fbs      9
restecg  9
thalach  9
exang    9
oldpeak  9
slope    9
ca       9
thal     9
target   9
dtype: int64

```

```

Q1 = df["trestbps"].quantile(0.25)
Q3 = df["trestbps"].quantile(0.75)
IQR = Q3 - Q1
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR
outliers = df[(df["trestbps"] < lower_bound) |
(df["trestbps"] > upper_bound)]
print(outliers)
print("Number of outliers:", len(outliers))

```

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	\
8	52	1	2	172	199	1	1	162	0	0.5	
101	59	1	3	178	270	0	0	145	0	4.2	
110	64	0	0	180	325	0	1	154	1	0.0	
203	68	1	2	180	274	1	0	150	1	1.6	
223	56	0	0	200	288	1	0	133	1	4.0	
241	59	0	0	174	249	0	1	143	1	0.0	
248	54	1	1	192	283	0	0	195	0	0.0	
260	66	0	0	178	228	1	1	165	1	1.0	
266	55	0	0	180	327	0	2	117	1	3.4	

	slope	ca	thal	target
8	2	0	3	1
101	0	0	3	1
110	2	0	2	1
203	1	0	3	0
223	0	2	3	0
241	1	0	2	0
248	2	1	3	0
260	1	2	3	0
266	1	0	2	0

Number of outliers: 9